Final Assessment — Introduction to Data Science

11372 & 11516

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```
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)

## ----------------------------------------------------------------------------
##  DO NOT EDIT personalise.R. Run this chunk once to build your dataset.
## ----------------------------------------------------------------------------
source("personalise.R")

my <- make_my_dataset(params$studentID,
                      raw_dir = "data_raw",
                      out_dir = "data")
```

```
## Student ID   : u3284342
## Seed         : 12109058
## Window       : 2004 - 2023
## Countries    : 150
## Files written to /home/ruhanshafi/Projects/IntroDS/Assessment2/data
##   Emissions.csv   3005 rows
##   Renewables.csv  148 rows x 21 cols
##   Consumption.csv 1897 rows
##   Countries.csv   150 rows
## FINGERPRINT  : EM-606669-0798-6584
```

```
# Packages used in this report. Install once with:
# install.packages(c("tidyverse", "lubridate", "scales", "knitr",
#                    "glmnet", "rpart"))
library(tidyverse)  # dplyr, tidyr, readr, stringr, forcats, ggplot2
library(lubridate)  # robust parsing of the inconsistent Year formats
library(scales)     # comma and log-scale axis labels
library(knitr)      # kable() tables
library(glmnet)     # ridge and lasso regression (Task 3.8)
library(rpart)      # regression tree (Task 3.8)

theme_set(theme_minimal(base_size = 11))
```

```
# Helper functions used throughout Task 1. Function names follow Google's
# R Style Guide (BigCamelCase); objects use snake_case.

# Convert text to numbers, tolerating thousands separators, currency
# symbols, stray spaces and the Unicode minus sign. Anything that still is
# not a number becomes NA, so it can be counted as a defect.
ToNumber <- function(x) {
  x <- str_replace_all(as.character(x), "−", "-")
  x <- str_remove_all(x, "[,$\\s]")
  suppressWarnings(as.numeric(x))
}

# Label each Year value by its format, so the variants can be reported.
ClassifyYearFormat <- function(x) {
  x <- str_trim(as.character(x))
```

```r
  case_when(
    is.na(x) | x == "" ~ "missing",
    str_detect(x, "^\\d{4}$") ~ "YYYY",
    str_detect(x, "^\\d{4}\\.0+$") ~ "YYYY.0 (decimal)",
    str_detect(x, "^\\d{4}-\\d{1,2}-\\d{1,2}") ~ "ISO date (YYYY-MM-DD)",
    str_detect(x, "^\\d{1,2}[/.-]\\d{1,2}[/.-]\\d{4}$") ~ "D/M/YYYY date",
    str_detect(x, "^\\d{5}$") ~ "Excel serial number",
    str_detect(x, "^'?\\d{2}$") ~ "two-digit year",
    str_detect(x, "[A-Za-z]") & str_detect(x, "\\d{4}") ~ "text + year",
    TRUE ~ "other"
  )
}

# Convert every Year variant to a four-digit integer year.
ParseYear <- function(x) {
  x <- str_trim(as.character(x))
  fmt <- ClassifyYearFormat(x)
  out <- rep(NA_integer_, length(x))

  idx <- fmt %in% c("YYYY", "YYYY.0 (decimal)", "ISO date (YYYY-MM-DD)")
  out[idx] <- as.integer(str_sub(x[idx], 1, 4))

  idx <- fmt == "D/M/YYYY date"
  out[idx] <- year(parse_date_time(x[idx], orders = c("dmy", "mdy")))

  idx <- fmt == "Excel serial number"
  out[idx] <- year(as_date(as.numeric(x[idx]), origin = "1899-12-30"))

  idx <- fmt == "two-digit year"
  yy <- as.integer(str_remove(x[idx], "'"))
  out[idx] <- if_else(yy > 50L, 1900L + yy, 2000L + yy)

  idx <- fmt == "text + year"
  out[idx] <- as.integer(str_extract(x[idx], "(19|20)\\d{2}"))

  idx <- fmt == "other"
  out[idx] <- year(parse_date_time(x[idx],
                                   orders = c("Ymd", "dmY", "mdY", "bY")))
  out
}

# Normalise country names so the same country matches across files
# (case, accents, punctuation, "&", leading "The").
NormaliseName <- function(x) {
  x %>%
    iconv(to = "ASCII//TRANSLIT", sub = "") %>%
    str_to_lower() %>%
    str_replace_all("&", "and") %>%
    str_remove_all("[^a-z0-9 ]") %>%
    str_remove("^the ") %>%
    str_squish()
}
```

```r
# Known spelling variants (normalised alias = normalised canonical name).
# DECISION: extend this list with any names printed as "Unmatched name -
# review" in Task 1.2.
name_fixes <- c(
  "united states of america" = "united states",
  "usa" = "united states",
  "us" = "united states",
  "russian federation" = "russia",
  "korea rep" = "south korea",
  "republic of korea" = "south korea",
  "viet nam" = "vietnam",
  "czech republic" = "czechia",
  "turkiye" = "turkey",
  "iran islamic rep" = "iran",
  "egypt arab rep" = "egypt",
  "uk" = "united kingdom",
  "slovak republic" = "slovakia",
  "kyrgyz republic" = "kyrgyzstan",
  "lao pdr" = "laos",
  "dr congo" = "democratic republic of congo",
  "congo dem rep" = "democratic republic of congo",
  "congo rep" = "congo",
  "ivory coast" = "cote divoire",
  "burma" = "myanmar",
  "syrian arab republic" = "syria",
  "venezuela bolivarian republic of" = "venezuela"
)

ApplyNameFixes <- function(key) {
  if_else(key %in% names(name_fixes), unname(name_fixes[key]), key)
}

# Linear interpolation within one country's series. End gaps take the
# nearest observed value (rule = 2). A series with a single observation is
# filled with that value; an all-NA series is returned unchanged.
InterpolateWithin <- function(year, value) {
  ok <- !is.na(value)
  if (sum(ok) == 0) return(value)
  if (sum(ok) == 1) return(replace(value, !ok, value[ok]))
  approx(year[ok], value[ok], xout = year, rule = 2)$y
}

# Flag values more than `multiple` times above (or below) the country's
# own median: the signature of a unit or keying error (e.g. x1000).
FlagSpikes <- function(value, multiple = 10) {
  med <- median(value[value > 0], na.rm = TRUE)
  if (is.na(med)) return(rep(FALSE, length(value)))
  high <- value > multiple * med
  low <- med > 1 & value > 0 & value < med / multiple
  replace(high | low, is.na(high | low), FALSE)
}

# Most frequent non-missing value (NA if there is none).
```

```r
ModeValue <- function(x) {
  counts <- table(x)
  if (length(counts) == 0) return(NA_character_)
  names(sort(counts, decreasing = TRUE))[1]
}

# A readable "Country Year" label for example rows in the audit table.
LabelRows <- function(df) {
  id <- if ("Country" %in% names(df)) df$Country else df$ISO
  if ("Year" %in% names(df)) paste(id, df$Year) else as.character(id)
}

# Defect log: every detection step below appends a row, which becomes the
# Data Quality Audit table in Task 1A.
defect_log <- tibble(
  File = character(), Column = character(), Rows = integer(),
  Example = character(), Detection = character(), Handling = character()
)

LogDefect <- function(log, file, column, rows_df, detection, handling) {
  examples <- if (nrow(rows_df) == 0) "-" else
    paste(head(unique(LabelRows(rows_df)), 2), collapse = "; ")
  add_row(log, File = file, Column = column, Rows = nrow(rows_df),
          Example = examples, Detection = detection, Handling = handling)
}

# Codes sometimes used for "missing" in place of NA.
sentinel_values <- c(-999, -9999, -99999, 99999, 999999)

# Model evaluation metrics (Task 3).
Rmse <- function(actual, predicted) sqrt(mean((actual - predicted)^2))
RSquared <- function(actual, predicted) {
  1 - sum((actual - predicted)^2) / sum((actual - mean(actual))^2)
}
```

Introduction

Understanding which countries are responsible for carbon dioxide emissions, and how that responsibility should be measured, is central to climate policy. This report examines national CO2 emissions and renewable electricity generation using data from Our World in Data's CO2 and energy datasets, which compile figures from the Global Carbon Project, Ember, the Energy Institute and the World Bank. Although the underlying values are genuine, the slice analysed here is personalised from my student ID: it covers 150 countries across six regions over the period 2004–2023 and arrives as four deliberately imperfect files. The master panel, 'Emissions.csv', records annual production-based CO2 and oil CO2 for each country; 'Renewables.csv' reports renewable generation in a wide, name-keyed format; 'Consumption.csv' provides trade-adjusted, consumption-based emissions for a subset of countries; and 'Countries.csv' supplies static 2022 population and GDP figures. The files differ in shape, key structure and coverage, and each contains undisclosed data-quality defects, so preparing them for analysis is a substantial task in its own right.

The analysis is guided by four objectives. The first is to detect, document and defensibly repair those defects so that the four sources can be merged into a single tidy country-year panel. The second is to explore which countries and regions account for the largest share of emissions, and how rankings shift between total, per-capita and consumption-based measures. The third is to assess how well a country's cumulative emissions can be predicted from its economic and demographic characteristics, and whether the assumptions underlying linear regression hold for these data. The fourth is to translate these findings into a dashboard design that communicates the energy transition to a public audience.

The work is carried out in R using a small set of established packages. The 'tidyverse' provides the core tools for importing, reshaping and summarising data ('readr', 'tidyr', 'dplyr' and 'stringr') and for visualisation ('ggplot2'); 'lubridate' parses the inconsistent year formats found in the source files; 'scales' formats axis labels and logarithmic scales; 'knitr' renders tables; and 'glmnet' and 'rpart' implement the ridge, lasso and regression-tree models used to extend the linear analysis.

The anticipated outcome is a fully reproducible analysis: a cleaned master data frame in which every defect is logged alongside its method of detection and its treatment; a set of rankings and visualisations that distinguish countries that emit the most in total from those that emit most intensively; regression models evaluated on a held-out test set, with their assumptions examined and their limitations made explicit; and a dashboard design grounded in these results.

The report is organised to follow these objectives. Task 1 imports, cleans, merges and extends the data, followed by a Data Quality Audit (1A). Task 2 explores the cleaned data, including a counterintuitive finding (2A) and a critique of one visualisation (2B). Task 3 develops and evaluates the predictive models, examines their assumptions (3A) and critiques the choice of predictor (3B). Task 4 presents the dashboard design, and the report concludes with a summary of findings, a verification table, a reproducibility appendix and a Decision Log.

Assumptions

1. **Emissions.csv defines the country set.** It is the master panel, so its 150 countries and its 2004–2023 window define the scope. Totals describe this sample, not the whole world; Australia, for example, is not among my 150 countries.
2. **Units are consistent across years and countries:** emissions in million tonnes of CO2 (Mt), renewables in terawatt-hours (TWh), GDP in millions of international dollars and GDP per capita in international dollars.
3. **`annual_co2_mt` is production-based fossil CO2.** It records emissions inside a country's borders and contains no land-use sinks, so a negative value is treated as an error, not a removal.
4. **Static country data apply to every year.** `Countries.csv` holds a single 2022 value per country, so per-capita measures divide cumulative totals by 2022 population.
5. **"Cumulative" means cumulative within the window.** `CumCO2` and the other running totals start from 2004, not from a country's first historical emissions.

6. **Missing is not zero.** A missing value is filled from the same country's neighbouring years where possible; a country with no consumption data is assumed trade-neutral (`ConsCO2 = CO2`).
7. **Aggregate rows are not countries.** Rows such as "World" or "European Union (27)" duplicate the countries they contain and are excluded.
8. **The Paris split follows the brief.** "Pre-Paris" covers years before 2016 and "Post-Paris" 2016 onwards (the agreement was adopted in December 2015).

Task 1: Data preparation and wrangling

1.1 Import the four CSV files

```
# Everything is read as text first. Numbers stored as text, odd Year
# formats and stray whitespace would otherwise be coerced silently, and the
# defects would be lost before they could be counted.
ReadRaw <- function(path) {
  read_csv(path, col_types = cols(.default = col_character()),
           na = c("", "NA"), trim_ws = FALSE)
}

emissions_raw   <- ReadRaw("data/Emissions.csv")
renewables_raw  <- ReadRaw("data/Renewables.csv")
consumption_raw <- ReadRaw("data/Consumption.csv")
countries_raw   <- ReadRaw("data/Countries.csv")

raw_files <- list(Emissions = emissions_raw, Renewables = renewables_raw,
                  Consumption = consumption_raw, Countries = countries_raw)

tibble(File = names(raw_files),
       Rows = map_int(raw_files, nrow),
       Columns = map_int(raw_files, ncol),
       ColumnNames = map_chr(raw_files, ~ paste(head(names(.x), 6),
                                                collapse = ", "))) %>%
  kable(caption = "Raw files as imported")
```

Table 1: Raw files as imported

File	Rows	Columns	ColumnNames
Emissions	3005	6	iso_code, location, region, year, annual_co2_mt, annual_oil_co2_mt
Renewables	148	21	Country.Region, 2004, 2005, 2006, 2007, 2008
Consumption	1897	3	Country Code, Year, Consumption CO2
Countries	150	5	countryCode, Country, popData2022, GDP, GDP/capita

1.2 Reshape `Renewables.csv` to one row per country per year

Remember: this file contains rows that are not countries.

```
# Year columns are any whose name contains a four-digit number
# (e.g. "2010", "X2010"); the remaining column(s) identify the row.
year_cols <- names(renewables_raw)[str_detect(names(renewables_raw),
                                              "\\d{4}")]
id_cols <- setdiff(names(renewables_raw), year_cols)
cat("Identifier column(s):", paste(id_cols, collapse = ", "), "\n")
```

```
## Identifier column(s): Country.Region
```

```r
cat("Year columns:", length(year_cols), "\n")
```

```
## Year columns: 20
```

```r
# Wide -> long: one row per country per year. Year stays as text here and
# is standardised in 1.4 together with the other files.
renewables_long <- renewables_raw %>%
  rename(Country = all_of(id_cols[1])) %>%
  pivot_longer(all_of(year_cols), names_to = "Year",
               values_to = "Renewables") %>%
  mutate(Country = str_squish(Country),
         NameKey = ApplyNameFixes(NormaliseName(Country)))

# Renewables.csv has no ISO code, so names are mapped to ISO codes using
# the names in Emissions.csv and Countries.csv.
name_lookup <- bind_rows(
  emissions_raw %>%
    transmute(ISO = str_to_upper(str_trim(iso_code)), Name = location),
  countries_raw %>%
    transmute(ISO = str_to_upper(str_trim(countryCode)), Name = Country)
) %>%
  filter(str_detect(ISO, "^[A-Z]{3}$")) %>%
  mutate(NameKey = ApplyNameFixes(NormaliseName(Name))) %>%
  distinct(NameKey, .keep_all = TRUE) %>%
  select(NameKey, ISO)

renewables_long <- renewables_long %>%
  left_join(name_lookup, by = "NameKey")

# Rows whose name matches no country are either aggregates (World,
# continents, income groups, "(EI)"-style source regions) or spelling
# variants that still need adding to name_fixes.
aggregate_pattern <- paste0(
  "world|income|oecd|union|\\(|total|other|non-|^africa|^asia|^europe|",
  "america|oceania|middle east|cis|ussr|g20|g7|opec|pacific|caribbean|",
  "eurasia|region"
)

renew_non_country <- renewables_long %>%
  filter(is.na(ISO)) %>%
  distinct(Country) %>%
  mutate(Type = if_else(str_detect(str_to_lower(Country), aggregate_pattern),
                        "Aggregate / region",
                        "Unmatched name - review"))

kable(count(renew_non_country, Type),
      caption = "Renewables.csv rows that are not countries")
```

Table 2: Renewables.csv rows that are not countries

Type	n
Aggregate / region	2

```
print(renew_non_country, n = Inf)
```

```
## # A tibble: 2 x 2
##   Country             Type
##   <chr>               <chr>
## 1 European Union (27) Aggregate / region
## 2 World               Aggregate / region
```

```
defect_log <- LogDefect(
  defect_log, "Renewables.csv", "Country",
  renewables_long %>% filter(is.na(ISO)),
  paste("Names with no match in the Emissions/Countries name lookup",
        "after normalisation; classified by keyword"),
  "Removed (aggregates double-count countries already in the data)"
)

# DECISION: drop every row that does not map to a country ISO code.
renewables_long <- renewables_long %>%
  filter(!is.na(ISO)) %>%
  select(ISO, Country, Year, Renewables)
```

What did you do with the non-country rows, and why?

Renewables.csv identifies each row by name only (‘Country.Region‘), with one column per year from 2004 to 2023. After pivoting it to one row per country per year, I mapped each name to an ISO code using the country names in Emissions.csv and Countries.csv. Names were normalised (case, accents, punctuation) and known spelling variants were corrected first, such as “United States of America” for “United States”. Every row matched a country except two: “World” and “European Union (27)”.

I removed both rows, which is 40 country-year observations (2 rows × 20 years). They are not countries but aggregates: “World” is the sum of every country’s generation, and “European Union (27)” is the sum of its 27 member states, several of which (e.g. France, Germany, Italy) already appear as individual rows. Keeping them would break the one-country-per-year structure of the master data frame. They would also double-count renewable generation when totals are summed across countries in Task 2. They also have no ISO code, so they could not be joined to Emissions.csv.

1.3 Rename the columns

```
emissions <- emissions_raw %>%
  rename(ISO = iso_code, Country = location, Region = region,
         Year = year, CO2 = annual_co2_mt, OilCO2 = annual_oil_co2_mt)

consumption <- consumption_raw %>%
  rename(ISO = `Country Code`, ConsCO2 = `Consumption CO2`)

countries <- countries_raw %>%
```

```r
  rename(ISO = countryCode, Population = popData2022,
         GDPCapita = `GDP/capita`)

# --- Type conversion, with defect detection before each change ----------

# ISO codes with whitespace or lower case break joins silently.
for (name in c("emissions", "consumption", "countries")) {
  df <- get(name)
  bad_iso <- df %>%
    filter(!is.na(ISO), !str_detect(ISO, "^[A-Z]{3}$"),
           str_detect(str_to_upper(str_trim(ISO)), "^[A-Z]{3}$"))
  file <- paste0(str_to_title(name), ".csv")
  defect_log <- LogDefect(
    defect_log, file, "ISO", bad_iso,
    paste("Failed the three-capital-letter regex but passed after",
          "str_trim() and str_to_upper()"),
    "Trimmed and upper-cased"
  )
}

# Values that are present but not numeric (e.g. "n/a", "unknown").
numeric_targets <- list(
  list("emissions", "Emissions.csv", c("CO2", "OilCO2")),
  list("consumption", "Consumption.csv", "ConsCO2"),
  list("countries", "Countries.csv", c("Population", "GDP", "GDPCapita")),
  list("renewables_long", "Renewables.csv", "Renewables")
)

for (target in numeric_targets) {
  df <- get(target[[1]])
  for (col in target[[3]]) {
    non_numeric <- df[!is.na(df[[col]]) & is.na(ToNumber(df[[col]])), ]
    defect_log <- LogDefect(
      defect_log, target[[2]], col, non_numeric,
      "Non-missing text that ToNumber() could not convert to a number",
      "Set to NA, then treated in Task 1.6"
    )
  }
}

# Numbers written with thousands separators (e.g. "9,443.433") are read
# as text; ToNumber() repairs them, but they are a formatting defect.
for (target in numeric_targets) {
  df <- get(target[[1]])
  for (col in target[[3]]) {
    comma_rows <- df[!is.na(df[[col]]) & str_detect(df[[col]], ","), ]
    defect_log <- LogDefect(
      defect_log, target[[2]], col, comma_rows,
      "str_detect(value, \",\"): thousands separator stored as text",
      "Separator removed by ToNumber() before conversion"
    )
  }
}
```

```r
# Region labels with inconsistent spacing or case.
region_variants <- emissions %>%
  filter(!is.na(Region), Region != str_to_title(str_squish(Region)))
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "Region", region_variants,
  "table(Region) showed the same continent spelt in more than one way",
  "str_squish() + str_to_title(), then each country's most common label"
)

# Now convert.
emissions <- emissions %>%
  mutate(ISO = str_to_upper(str_trim(ISO)),
         Country = str_squish(Country),
         Region = str_to_title(str_squish(Region)),
         across(c(CO2, OilCO2), ToNumber))

consumption <- consumption %>%
  mutate(ISO = str_to_upper(str_trim(ISO)), ConsCO2 = ToNumber(ConsCO2))

countries <- countries %>%
  mutate(ISO = str_to_upper(str_trim(ISO)),
         Country = str_squish(Country),
         across(c(Population, GDP, GDPCapita), ToNumber))

renewables_long <- renewables_long %>%
  mutate(Renewables = ToNumber(Renewables))

map(list(Emissions = emissions, Consumption = consumption,
         Countries = countries, Renewables = renewables_long), names)
```

```
## $Emissions
## [1] "ISO"     "Country" "Region"  "Year"    "CO2"     "OilCO2"
##
## $Consumption
## [1] "ISO"     "Year"    "ConsCO2"
##
## $Countries
## [1] "ISO"        "Country"    "Population" "GDP"        "GDPCapita"
##
## $Renewables
## [1] "ISO"        "Country"    "Year"       "Renewables"
```

1.4 Make Year consistent across all data frames

```r
# Detection: which Year formats appear in each file?
year_formats <- bind_rows(
  tibble(File = "Emissions.csv", Format = ClassifyYearFormat(emissions$Year)),
  tibble(File = "Consumption.csv",
         Format = ClassifyYearFormat(consumption$Year)),
  tibble(File = "Renewables.csv (column names)",
         Format = ClassifyYearFormat(renewables_long$Year))
```

```
)
year_formats %>%
  count(File, Format) %>%
  kable(caption = "Year formats found in each file")
```

Table 3: Year formats found in each file

File	Format	n
Consumption.csv	D/M/YYYY date	45
Consumption.csv	YYYY	1783
Consumption.csv	YYYY.0 (decimal)	38
Consumption.csv	text + year	31
Emissions.csv	YYYY	3005
Renewables.csv (column names)	YYYY	2920

```
for (name in c("emissions", "consumption")) {
  df <- get(name)
  odd <- df %>% filter(ClassifyYearFormat(Year) != "YYYY")
  defect_log <- LogDefect(
    defect_log, paste0(str_to_title(name), ".csv"), "Year", odd,
    "ClassifyYearFormat() found values not in plain YYYY form",
    "Converted to integer year with ParseYear() (lubridate for dates)"
  )
}

# Convert every Year to the same type: a four-digit integer.
emissions <- emissions %>% mutate(Year = ParseYear(Year))
consumption <- consumption %>% mutate(Year = ParseYear(Year))
renewables_long <- renewables_long %>% mutate(Year = ParseYear(Year))

# Check: same class everywhere, and no values that failed to parse.
tibble(
  Frame = c("emissions", "consumption", "renewables_long"),
  Class = c(class(emissions$Year), class(consumption$Year),
            class(renewables_long$Year)),
  Unparsed = c(sum(is.na(emissions$Year)), sum(is.na(consumption$Year)),
               sum(is.na(renewables_long$Year))),
  Range = c(paste(range(emissions$Year, na.rm = TRUE), collapse = "-"),
            paste(range(consumption$Year, na.rm = TRUE), collapse = "-"),
            paste(range(renewables_long$Year, na.rm = TRUE),
                  collapse = "-"))
) %>%
  kable(caption = "Year after standardisation")
```

Table 4: Year after standardisation

Frame	Class	Unparsed	Range
emissions	numeric	0	2004-2023
consumption	numeric	0	2004-2023
renewables_long	numeric	0	2004-2023

```r
# Rows whose Year could not be recovered cannot be placed in time.
emissions <- filter(emissions, !is.na(Year))
consumption <- filter(consumption, !is.na(Year))
renewables_long <- filter(renewables_long, !is.na(Year))
```

1.4b Remove non-country rows, duplicates and impossible values

```r
# --- Emissions.csv (the master panel) ------------------------------------
n_emissions_raw <- nrow(emissions)

# 1. Non-country rows (aggregates have codes such as OWID_WRL or no code).
emissions_non_country <- emissions %>%
  filter(is.na(ISO) | !str_detect(ISO, "^[A-Z]{3}$"))
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "ISO", emissions_non_country,
  "ISO missing or not three letters (aggregate rows such as OWID_*)",
  "Removed so totals are not double-counted"
)
emissions <- anti_join(emissions, emissions_non_country,
                       by = names(emissions))

# 2. Exact duplicate rows.
n_dup_exact <- nrow(emissions) - nrow(distinct(emissions))
dup_exact_rows <- emissions[duplicated(emissions), ]
emissions <- distinct(emissions)

# 3. Same country and year recorded twice with different values.
dup_key_rows <- emissions %>%
  group_by(ISO, Year) %>%
  filter(n() > 1) %>%
  ungroup()
emissions <- emissions %>%
  arrange(ISO, Year, is.na(CO2)) %>%   # keep the non-missing copy first
  distinct(ISO, Year, .keep_all = TRUE)
n_dup_key <- nrow(dup_key_rows) - n_distinct(dup_key_rows[c("ISO", "Year")])
n_dup_removed <- n_dup_exact + n_dup_key

defect_log <- LogDefect(
  defect_log, "Emissions.csv", "(whole row)", dup_exact_rows,
  "duplicated() on all columns", "Removed with distinct()"
)
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "ISO + Year", dup_key_rows,
  "group_by(ISO, Year) with n() > 1 after exact duplicates removed",
  "Kept the first non-missing record per country-year"
)
cat("Duplicate rows removed:", n_dup_removed,
    "(exact:", n_dup_exact, "| same key:", n_dup_key, ")\n")
```

```
## Duplicate rows removed: 5 (exact: 5 | same key: 0 )
```

```r
# 4. Years outside this student's data window (keying errors).
window_years <- range(as.integer(my$window))
out_of_window <- emissions %>%
  filter(Year < window_years[1] | Year > window_years[2])
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "Year", out_of_window,
  paste0("Year outside the personalised window ",
         window_years[1], "-", window_years[2]),
  "Removed"
)
emissions <- anti_join(emissions, out_of_window, by = c("ISO", "Year"))

# 5. One consistent name and region per country (most common label).
inconsistent_labels <- emissions %>%
  group_by(ISO) %>%
  filter(n_distinct(Country) > 1 | n_distinct(Region, na.rm = TRUE) > 1 |
           any(is.na(Region))) %>%
  ungroup()
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "Country / Region", inconsistent_labels,
  "group_by(ISO) with n_distinct(Country/Region) > 1 or missing Region",
  "Replaced with that country's most frequent label"
)
emissions <- emissions %>%
  group_by(ISO) %>%
  mutate(
    Country = ModeValue(Country),
    Region = ModeValue(Region)
  ) %>%
  ungroup()

# Keep an untouched copy of the values for the audit (1A).
emissions_before_fix <- emissions

# 6. Missing-value codes such as -999.
sentinel_rows <- emissions %>%
  filter(CO2 %in% sentinel_values | OilCO2 %in% sentinel_values)
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "CO2 / OilCO2", sentinel_rows,
  "Exact matches to codes like -999 / 99999",
  "Treated as missing (NA), interpolated in Task 1.6"
)
emissions <- emissions %>%
  mutate(across(c(CO2, OilCO2), ~ if_else(.x %in% sentinel_values,
                                          NA_real_, .x)))

# 7. Negative annual emissions (Task 1A ii).
negative_rows <- emissions %>%
  filter(CO2 < 0) %>%
  select(ISO, Country, Year, CO2)
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "CO2", negative_rows,
  "summary(CO2) showed a negative minimum; filter(CO2 < 0)",
```

```
  "Set to NA and interpolated from neighbouring years (Task 1.6)"
)
neg_oil <- emissions %>% filter(OilCO2 < 0)
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "OilCO2", neg_oil,
  "filter(OilCO2 < 0)", "Set to NA and interpolated (Task 1.6)"
)
# DECISION: negative values become NA and are interpolated, not set to 0.
emissions <- emissions %>%
  mutate(CO2 = if_else(CO2 < 0, NA_real_, CO2),
         OilCO2 = if_else(OilCO2 < 0, NA_real_, OilCO2))

# 8. Spikes: a value 10x away from the country's own median.
emissions <- emissions %>%
  group_by(ISO) %>%
  mutate(CO2Spike = FlagSpikes(CO2), OilSpike = FlagSpikes(OilCO2)) %>%
  ungroup()
spike_rows <- emissions %>% filter(CO2Spike | OilSpike)
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "CO2 / OilCO2", spike_rows,
  "Value more than 10x above/below the country's own median (FlagSpikes)",
  "Set to NA and interpolated (likely unit or keying error)"
)
emissions <- emissions %>%
  mutate(CO2 = if_else(CO2Spike, NA_real_, CO2),
         OilCO2 = if_else(OilSpike, NA_real_, OilCO2)) %>%
  select(-CO2Spike, -OilSpike)

# 9. Oil emissions larger than total emissions (logically impossible).
oil_gt_total <- emissions %>% filter(OilCO2 > CO2)
defect_log <- LogDefect(
  defect_log, "Emissions.csv", "OilCO2", oil_gt_total,
  "filter(OilCO2 > CO2): a component cannot exceed its total",
  "OilCO2 set to NA and interpolated"
)
emissions <- emissions %>%
  mutate(OilCO2 = if_else(!is.na(CO2) & OilCO2 > CO2, NA_real_, OilCO2))

# 10. Missing years inside a country's record (reported, not filled).
year_gaps <- emissions %>%
  group_by(ISO, Country) %>%
  summarise(YearsPresent = n(),
            YearsExpected = max(Year) - min(Year) + 1, .groups = "drop") %>%
  filter(YearsPresent < YearsExpected)
cat("Countries with missing years inside their record:", nrow(year_gaps),
    "\n")
```

```
## Countries with missing years inside their record: 0
```

```
# --- Consumption.csv -----------------------------------------------------
cons_dups <- consumption %>% group_by(ISO, Year) %>% filter(n() > 1) %>%
  ungroup()
defect_log <- LogDefect(
```

```
  defect_log, "Consumption.csv", "ISO + Year", cons_dups,
  "group_by(ISO, Year) with n() > 1", "Kept first non-missing record"
)
consumption <- consumption %>%
  arrange(ISO, Year, is.na(ConsCO2)) %>%
  distinct(ISO, Year, .keep_all = TRUE)

cons_bad <- consumption %>%
  filter(ConsCO2 %in% sentinel_values | ConsCO2 < 0)
defect_log <- LogDefect(
  defect_log, "Consumption.csv", "ConsCO2", cons_bad,
  "Missing-value codes or negative values (filter(ConsCO2 < 0))",
  "Set to NA and interpolated (Task 1.6)"
)
consumption <- consumption %>%
  mutate(ConsCO2 = if_else(ConsCO2 %in% sentinel_values | ConsCO2 < 0,
                           NA_real_, ConsCO2)) %>%
  group_by(ISO) %>%
  mutate(Spike = FlagSpikes(ConsCO2)) %>%
  ungroup()
defect_log <- LogDefect(
  defect_log, "Consumption.csv", "ConsCO2", filter(consumption, Spike),
  "Value more than 10x from the country's own median (FlagSpikes)",
  "Set to NA and interpolated"
)
consumption <- consumption %>%
  mutate(ConsCO2 = if_else(Spike, NA_real_, ConsCO2)) %>%
  select(-Spike)

# --- Countries.csv ------------------------------------------------------
ctry_dups <- countries %>% group_by(ISO) %>% filter(n() > 1) %>% ungroup()
defect_log <- LogDefect(
  defect_log, "Countries.csv", "ISO", ctry_dups,
  "group_by(ISO) with n() > 1 in a one-row-per-country table",
  "Kept the most complete record"
)
countries <- countries %>%
  arrange(ISO, is.na(Population) + is.na(GDP) + is.na(GDPCapita)) %>%
  distinct(ISO, .keep_all = TRUE)

ctry_bad <- countries %>%
  filter(Population <= 0 | GDP <= 0 | GDPCapita <= 0 |
           Population %in% sentinel_values | GDP %in% sentinel_values)
defect_log <- LogDefect(
  defect_log, "Countries.csv", "Population / GDP / GDPCapita", ctry_bad,
  "summary() showed zero/negative minima or missing-value codes",
  "Set to NA, then derived or imputed in Task 1.6"
)
countries <- countries %>%
  mutate(across(c(Population, GDP, GDPCapita),
                ~ if_else(.x <= 0 | .x %in% sentinel_values,
                          NA_real_, .x)))
```

```r
# The three columns are linked: GDP (millions) * 1e6 / Population should
# reproduce GDPCapita. When they disagree, the population implied by GDP
# and GDPCapita shows whether Population is the value that is wrong.
ctry_inconsistent <- countries %>%
  mutate(ImpliedPopulation = GDP * 1e6 / GDPCapita,
         Ratio = Population / ImpliedPopulation) %>%
  filter(!is.na(Ratio), abs(Ratio - 1) > 0.05)
print(select(ctry_inconsistent, ISO, Country, Population, GDP, GDPCapita,
             ImpliedPopulation, Ratio))
```

```
## # A tibble: 1 x 7
##   ISO   Country Population    GDP GDPCapita ImpliedPopulation    Ratio
##   <chr> <chr>        <dbl>  <dbl>     <dbl>             <dbl>    <dbl>
## 1 CHL   Chile        19553 448338     22929         19553317. 0.001000
```

```r
defect_log <- LogDefect(
  defect_log, "Countries.csv", "Population", ctry_inconsistent,
  paste("Population differs >5% from GDP * 1e6 / GDPCapita",
        "(ratio printed above)"),
  "Replaced with the population implied by GDP and GDPCapita"
)
# DECISION: GDP and GDPCapita agree with each other, so Population is the
# inconsistent value; replace it with the implied population.
countries <- countries %>%
  mutate(Population = if_else(ISO %in% ctry_inconsistent$ISO,
                              round(GDP * 1e6 / GDPCapita), Population))

# --- Renewables ------------------------------------------------------------
renew_dups <- renewables_long %>% group_by(ISO, Year) %>%
  filter(n() > 1) %>% ungroup()
defect_log <- LogDefect(
  defect_log, "Renewables.csv", "Country + Year", renew_dups,
  "Two name variants mapped to the same ISO and year",
  "Kept the first non-missing value"
)
renewables_long <- renewables_long %>%
  arrange(ISO, Year, is.na(Renewables)) %>%
  distinct(ISO, Year, .keep_all = TRUE)

renew_bad <- renewables_long %>%
  filter(Renewables < 0 | Renewables %in% sentinel_values)
defect_log <- LogDefect(
  defect_log, "Renewables.csv", "Renewables", renew_bad,
  "Negative generation or missing-value codes",
  "Set to NA and interpolated (Task 1.6)"
)
renewables_long <- renewables_long %>%
  mutate(Renewables = if_else(Renewables < 0 |
                                Renewables %in% sentinel_values,
                              NA_real_, Renewables))

# Blank cells inside the wide table (missing years for a country).
renew_blank <- renewables_long %>% filter(is.na(Renewables))
```

```r
defect_log <- LogDefect(
  defect_log, "Renewables.csv", "Renewables", renew_blank,
  "is.na(Renewables) after pivoting: empty cells in the wide table",
  "Interpolated from neighbouring years (Task 1.6)"
)
# DECISION: no spike check on renewables. Generation that grows from
# near zero (e.g. new solar programmes) legitimately exceeds 10x its
# median, so FlagSpikes() would remove real growth.

cat("Emissions rows: ", n_emissions_raw, "->", nrow(emissions), "\n")
```

```
## Emissions rows:  3005 -> 3000
```

```r
n_countries <- n_distinct(emissions$ISO)
cat("Countries in Emissions.csv:", n_countries, "\n")
```

```
## Countries in Emissions.csv: 150
```

1.5 Merge the five new variables into the master data frame

Add `Renewables`, `ConsCO2`, `Population`, `GDP`, `GDPCapita`.

```r
# Emissions is the master: left joins keep every country-year in it.
# Each source gets a match flag so a genuine NA value can be told apart
# from a row that failed to join.
#   Renewables  -> ISO + Year (ISO obtained from the country name in 1.2)
#   Consumption -> ISO + Year
#   Countries   -> ISO only (static 2022 reference values)
master <- emissions %>%
  left_join(renewables_long %>%
              select(ISO, Year, Renewables) %>%
              mutate(RenewMatch = TRUE),
            by = c("ISO", "Year")) %>%
  left_join(consumption %>%
              select(ISO, Year, ConsCO2) %>%
              mutate(ConsMatch = TRUE),
            by = c("ISO", "Year")) %>%
  left_join(countries %>%
              select(ISO, Population, GDP, GDPCapita) %>%
              mutate(CtryMatch = TRUE),
            by = "ISO")

# A left join must never add rows to the master.
stopifnot(nrow(master) == nrow(emissions))

join_report <- tibble(
  Join = c("Renewables (name -> ISO, + Year)", "Consumption (ISO + Year)",
           "Countries (ISO)"),
  MasterRowsUnmatched = c(sum(is.na(master$RenewMatch)),
                          sum(is.na(master$ConsMatch)),
                          sum(is.na(master$CtryMatch))),
  CountriesUnmatched = c(n_distinct(master$ISO[is.na(master$RenewMatch)]),
```

```
                        n_distinct(master$ISO[is.na(master$ConsMatch)]),
                        n_distinct(master$ISO[is.na(master$CtryMatch)])),
  SourceRowsUnused = c(
    nrow(anti_join(renewables_long, emissions, by = c("ISO", "Year"))),
    nrow(anti_join(consumption, emissions, by = c("ISO", "Year"))),
    nrow(anti_join(countries, emissions, by = "ISO"))
  )
)
kable(join_report, caption = "Rows that failed to match on each join")
```

Table 5: Rows that failed to match on each join

Join	MasterRowsUnmatched	CountriesUnmatched	SourceRowsUnused
Renewables (name -> ISO, + Year)	80	4	0
Consumption (ISO + Year)	1103	56	0
Countries (ISO)	0	0	0

```
# Which countries are behind the unmatched rows?
master %>%
  group_by(ISO, Country) %>%
  summarise(NoRenewables = all(is.na(RenewMatch)),
            NoConsumption = all(is.na(ConsMatch)),
            NoCountryInfo = all(is.na(CtryMatch)), .groups = "drop") %>%
  filter(NoRenewables | NoConsumption | NoCountryInfo) %>%
  print(n = 30)
```

```
## # A tibble: 56 x 5
##    ISO   Country                NoRenewables NoConsumption NoCountryInfo
##    <chr> <chr>                  <lgl>        <lgl>         <lgl>
##  1 ABW   Aruba                  FALSE        TRUE          FALSE
##  2 AFG   Afghanistan            FALSE        TRUE          FALSE
##  3 AGO   Angola                 FALSE        TRUE          FALSE
##  4 AND   Andorra                TRUE         TRUE          FALSE
##  5 ATG   Antigua and Barbuda    FALSE        TRUE          FALSE
##  6 BDI   Burundi                FALSE        TRUE          FALSE
##  7 BIH   Bosnia and Herzegovina FALSE        TRUE          FALSE
##  8 BLZ   Belize                 FALSE        TRUE          FALSE
##  9 BMU   Bermuda                FALSE        TRUE          FALSE
## 10 BRB   Barbados               FALSE        TRUE          FALSE
## 11 BTN   Bhutan                 FALSE        TRUE          FALSE
## 12 COG   Congo                  FALSE        TRUE          FALSE
## 13 COM   Comoros                FALSE        TRUE          FALSE
## 14 CUB   Cuba                   FALSE        TRUE          FALSE
## 15 DMA   Dominica               FALSE        TRUE          FALSE
## 16 DZA   Algeria                FALSE        TRUE          FALSE
## 17 ERI   Eritrea                FALSE        TRUE          FALSE
## 18 FJI   Fiji                   FALSE        TRUE          FALSE
## 19 GAB   Gabon                  FALSE        TRUE          FALSE
## 20 GNB   Guinea-Bissau          FALSE        TRUE          FALSE
## 21 GNQ   Equatorial Guinea      FALSE        TRUE          FALSE
## 22 GRD   Grenada                FALSE        TRUE          FALSE
```

```
## 23 GRL   Greenland              FALSE        TRUE          FALSE
## 24 GUY   Guyana                 FALSE        TRUE          FALSE
## 25 IRQ   Iraq                   FALSE        TRUE          FALSE
## 26 ISL   Iceland                FALSE        TRUE          FALSE
## 27 KNA   Saint Kitts and Nevis  FALSE        TRUE          FALSE
## 28 LBR   Liberia                FALSE        TRUE          FALSE
## 29 LBY   Libya                  FALSE        TRUE          FALSE
## 30 LIE   Liechtenstein          TRUE         TRUE          FALSE
## # i 26 more rows
```

```
master <- select(master, -RenewMatch, -ConsMatch, -CtryMatch)
```

How many rows failed to match on each join, and what did you do about them?

After the 5 duplicate rows were removed, the master data frame built from Emissions.csv contained 3,000 rows, covering 150 countries over 20 years. Each of the three sources was attached with a left join, so every master row was kept and no rows were added.

The Renewables join, which used the ISO codes mapped from country names in 1.2 together with Year, left 80 master rows unmatched. All of them belong to 4 countries that have no row in Renewables.csv (4 countries $\times$ 20 years). In the other direction, no renewables row was left unused, which shows that the name mapping in 1.2 successfully matched every remaining country.

The Consumption join, on ISO and Year, left 1,103 master rows from 56 countries unmatched. Most of these come from 55 countries that have no consumption-based data at all ($55 \times 20 = 1{,}100$ rows); the remaining 3 rows are years missing from one country's otherwise complete record. Again, no row of Consumption.csv went unused. This confirms that trimming and upper-casing the ISO codes (for example " gha" to "GHA") in 1.3 and standardising the Year formats in 1.4 fixed every join failure those defects would otherwise have caused. The Countries join, on ISO alone, matched every master row.

I kept the unmatched rows in the master rather than dropping them, because removing them would delete those countries' emissions from every total in Task 2. Instead, their missing values were filled in Task 1.6, and flag columns (RenewImputed and ConsImputed) record which countries' values were imputed so that they can be identified in later analysis.

1.6 Handle the NAs

```
na_before <- colSums(is.na(master))

master <- master %>%
  group_by(ISO) %>%
  arrange(Year, .by_group = TRUE) %>%
  mutate(
    # Record coverage before anything is filled in.
    ConsImputed = all(is.na(ConsCO2)),
    RenewImputed = all(is.na(Renewables)),
    # Annual series change smoothly, so gaps inside a country's record
    # (including the negative/spike values set to NA) are interpolated.
    CO2 = InterpolateWithin(Year, CO2),
    OilCO2 = InterpolateWithin(Year, OilCO2),
    Renewables = InterpolateWithin(Year, Renewables),
    ConsCO2 = InterpolateWithin(Year, ConsCO2)
  ) %>%
  ungroup() %>%
```

```r
  mutate(
    # A country whose CO2 is missing in every year cannot be recovered.
    CO2 = replace_na(CO2, 0),
    # DECISION: a country with no consumption data at all is assumed
    # trade-neutral (ConsCO2 = CO2, so NetImported = 0) rather than 0,
    # which would make it look like a huge carbon exporter (see 1A iii).
    ConsCO2 = if_else(is.na(ConsCO2), CO2, ConsCO2),
    # No renewable data at all: 0, flagged by RenewImputed.
    Renewables = replace_na(Renewables, 0),
    # Oil emissions cannot exceed the total after interpolation.
    OilCO2 = pmin(replace_na(OilCO2, 0), CO2)
  )

# Static variables: derive one from the other two where possible ...
master <- master %>%
  mutate(GDP = coalesce(GDP, GDPCapita * Population / 1e6),
         GDPCapita = coalesce(GDPCapita, GDP * 1e6 / Population))

# ... otherwise use the region median (one value per country, so large
# regions are not weighted by their number of rows).
region_medians <- master %>%
  distinct(ISO, Region, Population, GDPCapita) %>%
  group_by(Region) %>%
  summarise(MedPop = median(Population, na.rm = TRUE),
            MedGDPCapita = median(GDPCapita, na.rm = TRUE),
            .groups = "drop")

master <- master %>%
  left_join(region_medians, by = "Region") %>%
  mutate(PopImputed = is.na(Population),
         Population = coalesce(Population, MedPop),
         GDPCapita = coalesce(GDPCapita, MedGDPCapita),
         GDP = coalesce(GDP, GDPCapita * Population / 1e6)) %>%
  select(-MedPop, -MedGDPCapita)

tibble(Variable = names(na_before),
       NA_before = na_before,
       NA_after = colSums(is.na(master))[names(na_before)]) %>%
  filter(NA_before > 0) %>%
  kable(caption = "Missing values before and after treatment")
```

Table 6: Missing values before and after treatment

Variable	NA_before	NA_after
CO2	4	0
OilCO2	5	0
Renewables	84	0
ConsCO2	1105	0
GDP	500	0
GDPCapita	500	0

```r
cat("Countries with consumption imputed as CO2:",
    n_distinct(master$ISO[master$ConsImputed]), "\n")
```

```
## Countries with consumption imputed as CO2: 55
```

```r
cat("Countries with no renewables data (set to 0):",
    n_distinct(master$ISO[master$RenewImputed]), "\n")
```

```
## Countries with no renewables data (set to 0): 4
```

```r
cat("Countries with population from the region median:",
    n_distinct(master$ISO[master$PopImputed]), "\n")
```

```
## Countries with population from the region median: 0
```

Justify your treatment of each variable. Different variables may warrant different choices.

The 4 missing CO2 values are the negative figures removed in 1.4b, and the 5 missing OilCO2 values are -999 missing-value codes (for example Qatar 2016 and Zimbabwe 2010); each is listed in the defect log. I filled them by linear interpolation from the same country's neighbouring years. Annual emissions change gradually and each gap is a single year, so interpolation produces a realistic estimate, whereas replacing them with 0 would understate CumCO2 in every subsequent year.

Renewables had 84 missing values, which needed two different treatments. Four of them are Romania's blank cells for 2015 to 2018, which I interpolated between its 2014 value (27.12 TWh) and its 2019 value (24.63 TWh). The other 80 belong to the 4 countries with no renewables data at all. Because there is no country-specific value to interpolate from, I set these to 0 and flagged them with RenewImputed. I chose this over a regional average, which would invent generation that these countries may not have, while accepting that `Renew_PC` will be understated for any of them that do generate renewable electricity.

`ConsCO2` had the largest gap, with 1,105 missing values. For the 55 countries with no consumption data at all, I assumed trade neutrality by setting ConsCO2 equal to CO2, so their NetImported is 0, and recorded this with ConsImputed. Setting them to 0 instead would claim that these countries consume no emissions whatsoever, making NetImported equal to -CumCO2 and presenting every one of them as a large exporter of embodied carbon, a bias examined in 1A iii. The remaining gaps, the country missing 3 years and Panama's two implausible values (0.28 in 2007 and 0.17 in 2011, against a typical value of about 3.7), were interpolated within each country's own series.

GDP and GDPCapita each had 500 missing values, from 25 countries, such as Aruba and Andorra, that have neither figure, so one could not be derived from the other. I filled GDPCapita with the median of its region, calculated from one value per country so that regions were not weighted by their number of rows, and then calculated GDP as GDPCapita × Population / 1e6. This keeps each country's real population in its GDP estimate and keeps the three columns mathematically consistent. Dropping these countries would remove their emissions from every total, and using 0 would distort the correlations and models in Task 3.

Population had no missing values. Its only defect, Chile's recorded population of 19,553, had already been corrected in 1.4b to the value implied by its GDP and GDPCapita.

1.7 Add `Decade` and `Period`

```r
master <- master %>%
  mutate(
    Decade = paste0(Year %/% 10 * 10, "s"),
    Period = factor(if_else(Year < 2016, "Pre-Paris", "Post-Paris"),
                    levels = c("Pre-Paris", "Post-Paris"))
  )

count(master, Decade, Period)
```

```
## # A tibble: 4 x 3
##   Decade Period         n
##   <chr>  <fct>      <int>
## 1 2000s  Pre-Paris    900
## 2 2010s  Pre-Paris    900
## 3 2010s  Post-Paris   600
## 4 2020s  Post-Paris   600
```

1.8 Add the four cumulative variables

CumCO2, CumOilCO2, CumRenew, CumConsCO2.

```r
# Running totals must be computed within each country, in year order.
master <- master %>%
  arrange(ISO, Year) %>%
  group_by(ISO) %>%
  mutate(CumCO2 = cumsum(CO2),
         CumOilCO2 = cumsum(OilCO2),
         CumRenew = cumsum(Renewables),
         CumConsCO2 = cumsum(ConsCO2)) %>%
  ungroup()

# Check: one country's first rows.
master %>%
  filter(ISO == first(ISO)) %>%
  select(Country, Year, CO2, CumCO2, Renewables, CumRenew) %>%
  head(5)
```

```
## # A tibble: 5 x 6
##   Country  Year   CO2 CumCO2 Renewables CumRenew
##   <chr>   <dbl> <dbl>  <dbl>      <dbl>    <dbl>
## 1 Aruba    2004  2.62   2.62          0        0
## 2 Aruba    2005  2.72   5.33          0        0
## 3 Aruba    2006  2.71   8.04          0        0
## 4 Aruba    2007  2.82  10.9           0        0
## 5 Aruba    2008  2.66  13.5           0        0
```

1.9 Add `NetImported` and `OilShare`

```r
master <- master %>%
  mutate(NetImported = CumConsCO2 - CumCO2,
```

```
        # Guard against division by zero for a country with no emissions.
        OilShare = if_else(CumCO2 > 0, CumOilCO2 / CumCO2, 0))

summary(select(master, NetImported, OilShare))
```

```
##   NetImported          OilShare
##  Min.   :-24691.04   Min.   :0.07856
##  1st Qu.:     0.00   1st Qu.:0.36698
##  Median :     0.00   Median :0.62353
##  Mean   :   -30.04   Mean   :0.62215
##  3rd Qu.:    46.52   3rd Qu.:0.92137
##  Max.   :  7062.24   Max.   :1.00000
```

1.10 Add the four per-capita variables

CO2_PC, OilCO2_PC, Renew_PC, ConsCO2_PC.

```
# Emissions are in million tonnes, so * 1e6 / Population gives tonnes per
# person. For renewables (TWh), * 1e6 / Population gives TWh per million
# people.
master <- master %>%
  mutate(CO2_PC = CumCO2 * 1e6 / Population,
         OilCO2_PC = CumOilCO2 * 1e6 / Population,
         Renew_PC = CumRenew * 1e6 / Population,
         ConsCO2_PC = CumConsCO2 * 1e6 / Population)

summary(select(master, CO2_PC, OilCO2_PC, Renew_PC, ConsCO2_PC))
```

```
##      CO2_PC            OilCO2_PC            Renew_PC
##  Min.   :  0.01486   Min.   :  0.01434   Min.   :  0.00000
##  1st Qu.:  4.55992   1st Qu.:  2.68018   1st Qu.:  0.09512
##  Median : 21.41191   Median : 10.47611   Median :  1.35260
##  Mean   : 48.05527   Mean   : 23.08804   Mean   : 11.52225
##  3rd Qu.: 66.56134   3rd Qu.: 30.64318   3rd Qu.:  6.93466
##  Max.   :588.80658   Max.   :297.35973   Max.   :871.28781
##    ConsCO2_PC
##  Min.   :  0.01486
##  1st Qu.:  5.19373
##  Median : 23.12083
##  Mean   : 51.28262
##  3rd Qu.: 71.95324
##  Max.   :581.74870
```

```
# A quick look at the finished master data frame.
dim(master)
```

```
## [1] 3000   26
```

```
glimpse(master)
```

```
## Rows: 3,000
## Columns: 26
## $ ISO          <chr> "ABW", "ABW", "ABW", "ABW", "ABW", "ABW", "ABW", "ABW", "~
## $ Country      <chr> "Aruba", "Aruba", "Aruba", "Aruba", "Aruba", "Aruba", "Ar~
## $ Region       <chr> "North America", "North America", "North America", "North~
## $ Year         <dbl> 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 201~
## $ CO2          <dbl> 2.616, 2.715, 2.711, 2.821, 2.656, 2.627, 2.506, 2.495, 1~
## $ OilCO2       <dbl> 2.616, 2.715, 2.711, 2.821, 2.656, 2.627, 2.506, 2.495, 1~
## $ Renewables   <dbl> 0.00, 0.00, 0.00, 0.00, 0.00, 0.03, 0.11, 0.11, 0.14, 0.1~
## $ ConsCO2      <dbl> 2.616, 2.715, 2.711, 2.821, 2.656, 2.627, 2.506, 2.495, 1~
## $ Population   <dbl> 107792, 107792, 107792, 107792, 107792, 107792, 107792, 1~
## $ GDP          <dbl> 1194.874, 1194.874, 1194.874, 1194.874, 1194.874, 1194.87~
## $ GDPCapita    <dbl> 11085, 11085, 11085, 11085, 11085, 11085, 11085, 11085, 1~
## $ ConsImputed  <lgl> TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRU~
## $ RenewImputed <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, F~
## $ PopImputed   <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, F~
## $ Decade       <chr> "2000s", "2000s", "2000s", "2000s", "2000s", "2000s", "20~
## $ Period       <fct> Pre-Paris, Pre-Paris, Pre-Paris, Pre-Paris, Pre-Paris, Pr~
## $ CumCO2       <dbl> 2.616, 5.331, 8.042, 10.863, 13.519, 16.146, 18.652, 21.1~
## $ CumOilCO2    <dbl> 2.616, 5.331, 8.042, 10.863, 13.519, 16.146, 18.652, 21.1~
## $ CumRenew     <dbl> 0.00, 0.00, 0.00, 0.00, 0.00, 0.03, 0.14, 0.25, 0.39, 0.5~
## $ CumConsCO2   <dbl> 2.616, 5.331, 8.042, 10.863, 13.519, 16.146, 18.652, 21.1~
## $ NetImported  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ OilShare     <dbl> 1.0000000, 1.0000000, 1.0000000, 1.0000000, 1.0000000, 1.~
## $ CO2_PC       <dbl> 24.26896245, 49.45636040, 74.60664984, 100.77742319, 125.~
## $ OilCO2_PC    <dbl> 24.26896245, 49.45636040, 74.60664984, 100.77742319, 125.~
## $ Renew_PC     <dbl> 0.00000000, 0.00000000, 0.00000000, 0.00000000, 0.0000000~
## $ ConsCO2_PC   <dbl> 24.26896245, 49.45636040, 74.60664984, 100.77742319, 125.~
```

```
cat("Remaining NAs:", sum(is.na(master)), "\n")
```

```
## Remaining NAs: 0
```

1A. Data Quality Audit Report

*Approximately 250 words in total across the three parts below. Every claim must reference a specific value, country, year or column from **your** data.*

i. Defect inventory

```
# Evidence used for detection: summary() of the numeric columns before any
# fix (impossible minima, extreme maxima) ...
summary(select(emissions_before_fix, CO2, OilCO2))
```

```
##       CO2                OilCO2
##  Min.   :   -3.645   Min.   :-999.000
##  1st Qu.:    2.665   1st Qu.:   1.904
##  Median :   12.669   Median :   7.096
##  Mean   :  202.112   Mean   :  59.676
##  3rd Qu.:   71.999   3rd Qu.:  32.213
##  Max.   :12172.009   Max.   :2584.130
```

```
summary(select(countries_raw %>%
                 mutate(across(c(popData2022, GDP, `GDP/capita`),
                               ToNumber)),
               popData2022, GDP, `GDP/capita`))
```

```
##   popData2022             GDP             GDP/capita
##  Min.   :4.476e+03   Min.   :     675   Min.   :   689
##  1st Qu.:2.433e+06   1st Qu.:   36035   1st Qu.:  4868
##  Median :9.678e+06   Median :  145432   Median : 12659
##  Mean   :4.759e+07   Mean   :  937358   Mean   : 19442
##  3rd Qu.:3.303e+07   3rd Qu.:  504742   3rd Qu.: 26334
##  Max.   :1.425e+09   Max.   :26966017   Max.   :131991
##                      NAs    :25         NAs    :25
```

```
# ... and the defect log built step by step in Task 1 (detection and
# handling text is shown in full in the table below).
defect_log %>%
  filter(Rows > 0) %>%
  select(File, Column, Rows, Example) %>%
  kable(caption = "Defect log (generated from the detection code)")
```

Table 7: Defect log (generated from the detection code)

File	Column	Rows	Example
Renewables.csv	Country	40	European Union (27) 2004; European Union (27) 2005
Consumption.csv	ISO	76	are 2014; arg 2022
Consumption.csv	ConsCO2	5	CHN 2018; deu 2008
Consumption.csv	Year	114	ARE FY2004; ARE 01/01/2009
Emissions.csv	(whole row)	5	Maldives 2008; Sierra Leone 2016
Emissions.csv	CO2 / OilCO2	5	Bosnia and Herzegovina 2007; North Macedonia 2013

File	Column	Rows	Example
Emissions.csv	CO2	4	Equatorial Guinea 2010; Sri Lanka 2012
Consumption.csv	ConsCO2	2	PAN 2007; PAN 2011
Countries.csv	Population	1	Chile
Renewables.csv	Renewables	4	Romania 2015; Romania 2016

File	Column	Rows affected	Example (country, year)	How I detected it	How I handled it
Emissions.csv	All columns	5	Maldives 2008; Sierra Leone 2016	`duplicated()` across every column returned TRUE	Removed with `distinct()`
Emissions.csv	`annual_co2_mt` (CO2)	4	Sri Lanka 2012 (-3.645); Sweden 2022 (-2.594)	`summary()` showed a minimum of -3.645, which is physically impossible; `filter(CO2 < 0)` listed the rows	Set to NA, then interpolated from neighbouring years
Emissions.csv	`annual_oil_co2_mt` (OilCO2)	5	Bosnia and Herzegovina 2007; Qatar 2016 (-999)	`summary()` showed a minimum of -999; exact match to the missing-value code	Set to NA, then interpolated
Consumption.csv	`Country Code` (ISO)	76 (50 countries)	” are” (United Arab Emirates) 2014	Failed the three-capital-letter regex but passed after `str_trim()` + `str_to_upper()`	Trimmed and upper-cased
Consumption.csv	`Year`	114	United Arab Emirates “FY2004”, “01/01/2009”, “2010.0”	`ClassifyYearFormat()` count table showed four formats	Converted to integer with `ParseYear()` (lubridate for dates)
Consumption.csv	`Consumption CO2`	5	China 2018 (“9,443.433”)	`str_detect(value, ","):` thousands separator stored as text	Separator removed by `ToNumber()`
Consumption.csv	`Consumption CO2`	2	Panama 2007 (0.28); Panama 2011 (0.17)	`FlagSpikes()`: below one-tenth of Panama’s median (3.718)	Set to NA, then interpolated

File	Column	Rows affected	Example (country, year)	How I detected it	How I handled it
Consumption.csv	`Year` (missing years)	3	Panama 2006, 2008, 2016	Join report: one country with 3 unmatched country-years	Interpolated within Panama's series
Consumption.csv	Whole file (coverage)	55 countries (1,100 master rows)	Aruba, Afghanistan, Angola (all years)	Join report: no Consumption row for these ISO codes	`ConsCO2 = CO2`, flagged `ConsImputed` (see iii)
Countries.csv	`popData2022` (Population)	1	Chile 2022 (19,553)	`Population` differed from `GDP * 1e6 / GDPCapita` (ratio 0.001)	Replaced with implied population (19,553,317)
Countries.csv	`GDP`, `GDP/capita`	25 countries (500 master rows)	Aruba 2022; Andorra 2022	NA count in the 1.6 missing-value table	Region-median GDPCapita; GDP derived from it
Renewables.csv	`Country.Region`	2 rows (40 country-years)	World 2004; European Union (27) 2004	No match in the name lookup; aggregate keyword	Removed (would double-count countries)
Renewables.csv	`Country.Region`	6 names (120 country-years)	United States of America 2004; Burma 2004	Name absent from Emissions `location` after normalisation	Mapped with `name_fixes`
Renewables.csv	Year columns	4	Romania 2015–2018	`is.na(Renewables)` after pivoting: blank cells	Interpolated between 2014 (27.12) and 2019 (24.63)
Renewables.csv	Coverage	4 countries (80 master rows)	See the 1.5 join report	Join report: 80 master rows unmatched	Set to 0, flagged `RenewImputed`

Most defects were mechanical (duplicates, formats, keys, separators) and could be repaired without loss. The substantive ones (negative CO2, -999 codes, Chile's population) were found by testing values against physical or arithmetic limits, not by inspection.

ii. The negative emissions problem

```
# The negative rows, the value before treatment, and the value now in the
# master after interpolation from neighbouring years.
negative_rows %>%
  rename(CO2_original = CO2) %>%
  left_join(select(master, ISO, Year, CO2_imputed = CO2),
            by = c("ISO", "Year")) %>%
  kable(caption = "Negative annual CO2 rows", digits = 2)
```

Table 8: Negative annual CO2 rows

ISO	Country	Year	CO2_original	CO2_imputed
GNQ	Equatorial Guinea	2010	-1.00	5.47
LKA	Sri Lanka	2012	-3.64	14.31
SDN	Sudan	2018	-1.26	21.16
SWE	Sweden	2022	-2.59	37.64

```
# Context: two years either side of each negative value.
negative_rows %>%
  select(ISO, NegYear = Year) %>%
  inner_join(select(emissions_before_fix, ISO, Country, Year, CO2),
             by = "ISO", relationship = "many-to-many") %>%
  filter(abs(Year - NegYear) <= 2) %>%
  select(Country, Year, CO2) %>%
  kable(caption = "Neighbouring years", digits = 2)
```

Table 9: Neighbouring years

Country	Year	CO2
Equatorial Guinea	2008	5.93
Equatorial Guinea	2009	4.88
Equatorial Guinea	2010	-1.00
Equatorial Guinea	2011	6.07
Equatorial Guinea	2012	6.88
Sri Lanka	2010	12.70
Sri Lanka	2011	14.48
Sri Lanka	2012	-3.64
Sri Lanka	2013	14.14
Sri Lanka	2014	17.08
Sudan	2016	19.74
Sudan	2017	20.31
Sudan	2018	-1.26
Sudan	2019	22.01
Sudan	2020	19.80
Sweden	2020	36.62
Sweden	2021	38.57
Sweden	2022	-2.59
Sweden	2023	36.71

Which rows are they? Why do negative values legitimately appear in published emissions series? What did you do, and what alternative did you reject?

Four rows in Emissions.csv record negative annual emissions: Equatorial Guinea in 2010 (-0.995 Mt), Sri Lanka in 2012 (-3.645 Mt), Sudan in 2018 (-1.256 Mt) and Sweden in 2022 (-2.594 Mt). Negative values do appear legitimately in published series in two situations. The first is a genuine removal of carbon from the atmosphere, for example when a country's forests or other land-use sinks absorb more CO2 than it emits, giving negative net emissions. The second is an accounting revision: when a previously published estimate is corrected downward, a series recording annual changes or corrections can show a negative entry even though nothing was physically removed. Neither applies here. annual_co2_mt measures production-based emissions, which contain no sinks, and each negative value sits between ordinary positive years; Sweden, for

example, emitted 38.568 Mt in 2021 and 36.709 Mt in 2023. These rows are therefore data errors. I set them to NA and interpolated from the neighbouring years, giving Sweden 2022 a value of about 37.64 Mt. This keeps each country's cumulative total realistic. I rejected taking the absolute value: it would replace Sweden's 2022 figure with 2.594 Mt, still around 93% below its neighbours, so flipping the sign corrects the sign but not the size.

iii. Coverage gap in consumption-based data

```
no_cons <- master %>%
  filter(ConsImputed) %>%
  distinct(ISO, Country)
n_no_cons <- nrow(no_cons)
cat("Countries with no consumption-based data:", n_no_cons, "of",
    n_countries, "\n")
```

```
## Countries with no consumption-based data: 55 of 150
```

```
cat("Examples:", paste(head(no_cons$Country, 3), collapse = ", "), "\n")
```

```
## Examples: Aruba, Afghanistan, Angola
```

```
last_year_preview <- filter(master, Year == max(Year))

# Do covered and uncovered countries differ?
last_year_preview %>%
  group_by(Coverage = if_else(ConsImputed, "No consumption data",
                              "Has consumption data")) %>%
  summarise(Countries = n(),
            MedianPopulation_m = median(Population) / 1e6,
            MedianGDPCapita = median(GDPCapita),
            MedianCumCO2 = median(CumCO2),
            MedianCO2_PC = median(CO2_PC),
            .groups = "drop") %>%
  kable(digits = 1, caption = "Covered vs uncovered countries")
```

Table 10: Covered vs uncovered countries

Coverage	Countries	MedianPopulation_m	MedianGDPCapita	MedianCumCO2	MedianCO2_PC
Has consumption data	95	16.1	17023	821.7	78.4
No consumption data	55	2.1	11085	32.1	46.9

```
# What real covered countries look like: NetImported as a share of CumCO2.
last_year_preview %>%
  filter(!ConsImputed) %>%
  summarise(MinShare = min(NetImported / CumCO2),
            MedianShare = median(NetImported / CumCO2),
            MaxShare = max(NetImported / CumCO2)) %>%
  kable(digits = 3,
        caption = "NetImported / CumCO2 among countries with data")
```

Table 11: NetImported / CumCO2 among countries with data

MinShare	MedianShare	MaxShare
-0.49	0.15	2.572

```
# Counterfactual: what zero imputation would have produced instead.
# Zero gives NetImported = -CumCO2 (a share of -1) and ConsCO2_PC = 0.
zero_bias <- last_year_preview %>%
  filter(ConsImputed) %>%
  transmute(Country, CumCO2, CO2_PC,
            NetImported_zero = 0 - CumCO2,
            ConsCO2_PC_zero = 0,
            ConsCO2_PC_now = ConsCO2_PC) %>%
  arrange(NetImported_zero)

kable(head(zero_bias, 5), digits = 1,
      caption = "Most biased countries under zero imputation")
```

Table 12: Most biased countries under zero imputation

Country	CumCO2	CO2_PC	NetImported_zero	ConsCO2_PC_zero	ConsCO2_PC_now
Algeria	2898.4	63.7	-2898.4	0	63.7
Iraq	2889.6	65.6	-2889.6	0	65.6
Uzbekistan	2305.9	66.0	-2305.9	0	66.0
Libya	1141.4	158.0	-1141.4	0	158.0
Serbia	927.2	136.5	-927.2	0	136.5

How many countries have no coverage? Name three. Is imputing zero valid? If it biases `ConsCO2_PC` and `NetImported`, in which direction, and for which type of country is the bias worst? Support this with a number.

55 of the 150 countries in my Emissions.csv (37%) have no consumption-based data at all, for example Aruba, Afghanistan and Angola.

Imputing zero is not a valid strategy. A value of zero claims that a country's population consumed no goods or energy with embodied emissions, which is impossible. The data are missing, not zero. The coverage gap is also not random: the uncovered countries have a median population of 2.1 million against 16.1 million, a median GDP per capita of 11,085 against 17,023, and a median CumCO2 of 32.1 Mt against 821.7 Mt. A zero would therefore distort one distinct group of countries rather than add random noise.

Zero imputation biases both variables downward. ConsCO2_PC falls to 0 for every uncovered country, and NetImported becomes -CumCO2, a NetImported/CumCO2 share of -1. That is twice as extreme as the largest real net exporter among covered countries (a share of -0.49), and far below their median of +0.15, which shows that the typical covered country is a net importer. Every uncovered country would wrongly appear as an extreme exporter of embodied carbon.

The bias is worst for uncovered countries with large production-based emissions, which are typically fossil-fuel producers. In absolute terms, NetImported would be understated by the full cumulative total: 2,898.4 Mt for Algeria and 2,889.6 Mt for Iraq. Per person, ConsCO2_PC would fall from 158.0 t to 0 for Libya and from 136.5 t to 0 for Serbia. I therefore imputed ConsCO2 = CO2 (trade-neutral, NetImported = 0) instead, while recognising that, if the covered countries' median share of +0.15 also holds for these countries, this still slightly understates their consumption.

Task 2: Exploratory data analysis

2.1 Highest-emitting year, and the largest year-on-year fall

```
yearly_totals <- master %>%
  group_by(Year) %>%
  summarise(TotalCO2 = sum(CO2), Countries = n_distinct(ISO),
            .groups = "drop") %>%
  arrange(Year) %>%
  mutate(Change = TotalCO2 - lag(TotalCO2),
         PctChange = 100 * Change / lag(TotalCO2))

peak_year <- slice_max(yearly_totals, TotalCO2, n = 1, with_ties = FALSE)
largest_fall <- slice_min(yearly_totals, Change, n = 1, with_ties = FALSE)

cat("Highest total CO2:", peak_year$Year, "with",
    comma(peak_year$TotalCO2, accuracy = 0.1), "Mt\n")
```

```
## Highest total CO2: 2023 with 34,067.6 Mt
```

```
cat("Largest year-on-year fall:", largest_fall$Year, "(",
    comma(largest_fall$Change, accuracy = 0.1), "Mt,",
    round(largest_fall$PctChange, 2), "%)\n")
```

```
## Largest year-on-year fall: 2020 ( -1,329.5 Mt, -4.04 %)
```

```
# The Countries column confirms the fall is not caused by countries
# dropping out of the record that year.
kable(yearly_totals, digits = 1)
```

Year	TotalCO2	Countries	Change	PctChange
2004	24908.2	150	NA	NA
2005	25789.7	150	881.4	3.5
2006	26705.7	150	916.1	3.6
2007	27491.4	150	785.6	2.9
2008	28052.0	150	560.7	2.0
2009	27680.2	150	-371.8	-1.3
2010	29406.6	150	1726.4	6.2
2011	30495.7	150	1089.1	3.7
2012	30941.4	150	445.7	1.5
2013	31329.1	150	387.7	1.3
2014	31523.2	150	194.0	0.6
2015	31356.2	150	-166.9	-0.5
2016	31301.6	150	-54.7	-0.2
2017	31765.5	150	464.0	1.5
2018	32499.8	150	734.3	2.3
2019	32871.1	150	371.2	1.1
2020	31541.5	150	-1329.5	-4.0
2021	33080.4	150	1538.9	4.9

Year	TotalCO2	Countries	Change	PctChange
2022	33616.6	150	536.2	1.6
2023	34067.6	150	450.9	1.3

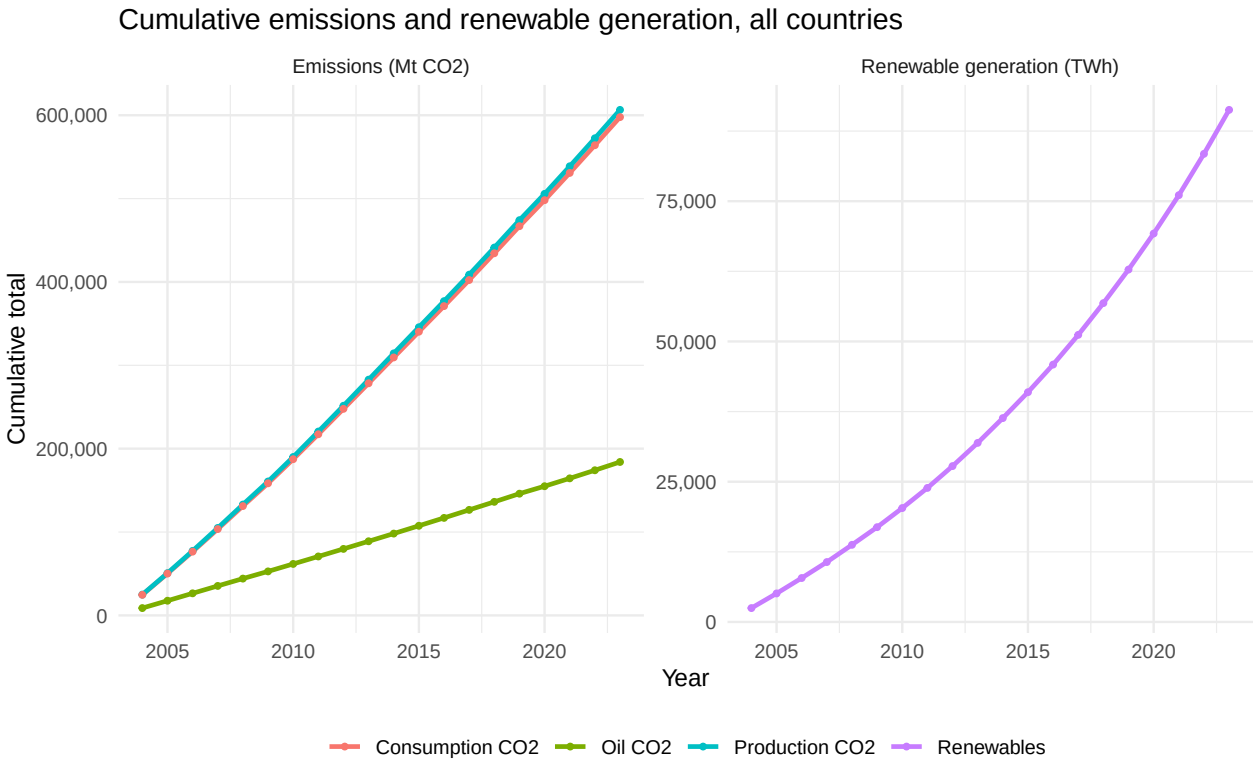
What caused the largest fall?

Total emissions peaked at 34,067.6 Mt in 2023, and the largest fall was in 2020 (-1,329.5 Mt, -4.04%), which I attribute to the COVID-19 pandemic, when lockdowns sharply cut transport and industrial fuel use. The fall is not a coverage artefact, since all 150 countries report in every year, and its temporary nature fits a pandemic shock: emissions rebounded by 1,538.9 Mt in 2021, the largest rise after 2010's recovery from the only other notable fall (-371.8 Mt in 2009, during the global financial crisis).

2.2 Cumulative totals over time for the whole dataset

```
cum_totals <- master %>%
  group_by(Year) %>%
  summarise(across(c(CumCO2, CumOilCO2, CumConsCO2, CumRenew), sum),
            .groups = "drop") %>%
  pivot_longer(-Year, names_to = "Measure", values_to = "Value") %>%
  mutate(
    # Emissions (Mt) and renewables (TWh) have different units, so they
    # get separate panels rather than one misleading shared axis.
    Panel = if_else(Measure == "CumRenew", "Renewable generation (TWh)",
                    "Emissions (Mt CO2)"),
    Measure = recode(Measure, CumCO2 = "Production CO2",
                     CumOilCO2 = "Oil CO2",
                     CumConsCO2 = "Consumption CO2",
                     CumRenew = "Renewables")
  )

ggplot(cum_totals, aes(Year, Value, colour = Measure)) +
  geom_line(linewidth = 1) +
  geom_point(size = 1) +
  facet_wrap(~ Panel, scales = "free_y") +
  scale_y_continuous(labels = label_comma()) +
  labs(title = "Cumulative emissions and renewable generation, all countries",
       x = "Year", y = "Cumulative total", colour = NULL) +
  theme(legend.position = "bottom")
```



Observation: by 2023 the 150 countries had emitted 606,423.7 Mt of production-based CO2 since 2004, with consumption-based emissions (597,745.8 Mt) tracking almost the same path, which is expected since trade only moves emissions between countries. Oil accounts for 30.3% (184,040.5 Mt) of the total. The renewables curve bends upward, unlike the near-straight emissions lines, showing that annual renewable generation grew faster over the period than annual emissions.

2.3 Extract `lastYear_data`

Compute the final year — do not type it in.

```
lastYear_data <- master %>% filter(Year == max(Year))

cat("Final year:", unique(lastYear_data$Year), "| Countries:",
    nrow(lastYear_data), "\n")
```

```
## Final year: 2023 | Countries: 150
```

2.4 Top 10 countries on four measures

top10total, top10percap, top10net, top10renew.

```
top10total <- lastYear_data %>%
  slice_max(CumCO2, n = 10, with_ties = FALSE) %>%
  select(ISO, Country, Region, CumCO2, Population, CO2_PC)

top10percap <- lastYear_data %>%
  slice_max(CO2_PC, n = 10, with_ties = FALSE) %>%
  select(ISO, Country, Region, CO2_PC, CumCO2, Population)
```

```r
top10net <- lastYear_data %>%
  slice_max(NetImported, n = 10, with_ties = FALSE) %>%
  select(ISO, Country, Region, NetImported, CumConsCO2, CumCO2)

top10renew <- lastYear_data %>%
  slice_max(Renew_PC, n = 10, with_ties = FALSE) %>%
  select(ISO, Country, Region, Renew_PC, CumRenew, Population)

kable(top10total, digits = 1, format.args = list(big.mark = ","),
      caption = "Top 10 by cumulative CO2 (Mt)")
```

Table 13: Top 10 by cumulative CO2 (Mt)

ISO	Country	Region	CumCO2	Population	CO2_PC
CHN	China	Asia	184,474.9	1,425,179,562	129.4
USA	United States	North America	109,445.5	341,534,041	320.5
IND	India	Asia	40,844.0	1,425,423,209	28.7
RUS	Russia	Europe	32,903.1	145,579,889	226.0
JPN	Japan	Asia	23,820.7	124,997,585	190.6
DEU	Germany	Europe	15,630.9	84,086,227	185.9
IRN	Iran	Asia	12,437.8	89,524,247	138.9
KOR	South Korea	Asia	11,848.1	51,782,515	228.8
SAU	Saudi Arabia	Asia	11,495.6	32,175,351	357.3
IDN	Indonesia	Asia	10,360.2	278,830,527	37.2

```r
kable(top10percap, digits = 1, format.args = list(big.mark = ","),
      caption = "Top 10 by cumulative CO2 per capita (t)")
```

Table 14: Top 10 by cumulative CO2 per capita (t)

ISO	Country	Region	CO2_PC	CumCO2	Population
QAT	Qatar	Asia	588.8	1,703.1	2,892,466
TTO	Trinidad and Tobago	North America	556.3	832.2	1,495,923
BHR	Bahrain	Asia	403.5	618.8	1,533,460
BRN	Brunei	Asia	389.7	177.4	455,374
ARE	United Arab Emirates	Asia	364.4	3,732.7	10,242,085
SAU	Saudi Arabia	Asia	357.3	11,495.6	32,175,351
USA	United States	North America	320.5	109,445.5	341,534,041
LUX	Luxembourg	Europe	305.7	199.7	653,315
ABW	Aruba	North America	297.4	32.1	107,792
NCL	New Caledonia	Oceania	293.4	84.2	287,130

```r
kable(top10net, digits = 1, format.args = list(big.mark = ","),
      caption = "Top 10 net importers of embodied CO2 (Mt)")
```

Table 15: Top 10 net importers of embodied CO2 (Mt)

ISO	Country	Region	NetImported	CumConsCO2	CumCO2
USA	United States	North America	7,062.2	116,507.7	109,445.5
JPN	Japan	Asia	3,566.0	27,386.6	23,820.7
GBR	United Kingdom	Europe	3,477.8	12,414.9	8,937.0
DEU	Germany	Europe	3,005.7	18,636.6	15,630.9
FRA	France	Europe	2,609.6	9,582.3	6,972.6
ITA	Italy	Europe	2,442.6	10,366.2	7,923.6
SGP	Singapore	Asia	2,366.7	3,286.8	920.1
HKG	Hong Kong	Asia	1,542.4	2,356.1	813.7
KOR	South Korea	Asia	1,512.7	13,360.8	11,848.1
ARE	United Arab Emirates	Asia	755.5	4,488.1	3,732.7

```
kable(top10renew, digits = 2, format.args = list(big.mark = ","),
      caption = "Top 10 by renewables per capita (TWh per million people)")
```

Table 16: Top 10 by renewables per capita (TWh per million people)

ISO	Country	Region	Renew_PC	CumRenew	Population
ISL	Iceland	Europe	871.29	331.41	380,368
NOR	Norway	Europe	499.84	2,727.54	5,456,796
BTN	Bhutan	Asia	189.90	148.30	780,921
SWE	Sweden	Europe	174.51	1,830.19	10,487,335
NZL	New Zealand	Oceania	128.93	661.63	5,131,731
GRL	Greenland	North America	122.11	6.84	56,014
AUT	Austria	Europe	106.69	967.13	9,064,678
FIN	Finland	Europe	105.16	585.67	5,569,300
URY	Uruguay	South America	59.83	202.87	3,390,913
DNK	Denmark	Europe	58.37	344.58	5,902,901

2.5 Totals by region

```
region_totals <- lastYear_data %>%
  group_by(Region) %>%
  summarise(Countries = n(),
            CumCO2 = sum(CumCO2),
            CumOilCO2 = sum(CumOilCO2),
            CumRenew = sum(CumRenew),
            CumConsCO2 = sum(CumConsCO2),
            .groups = "drop") %>%
  arrange(desc(CumCO2))

kable(region_totals, digits = 1, format.args = list(big.mark = ","),
      caption = "Cumulative totals by region (CO2 in Mt, renewables in TWh)")
```

Table 17: Cumulative totals by region (CO2 in Mt, renewables in TWh)

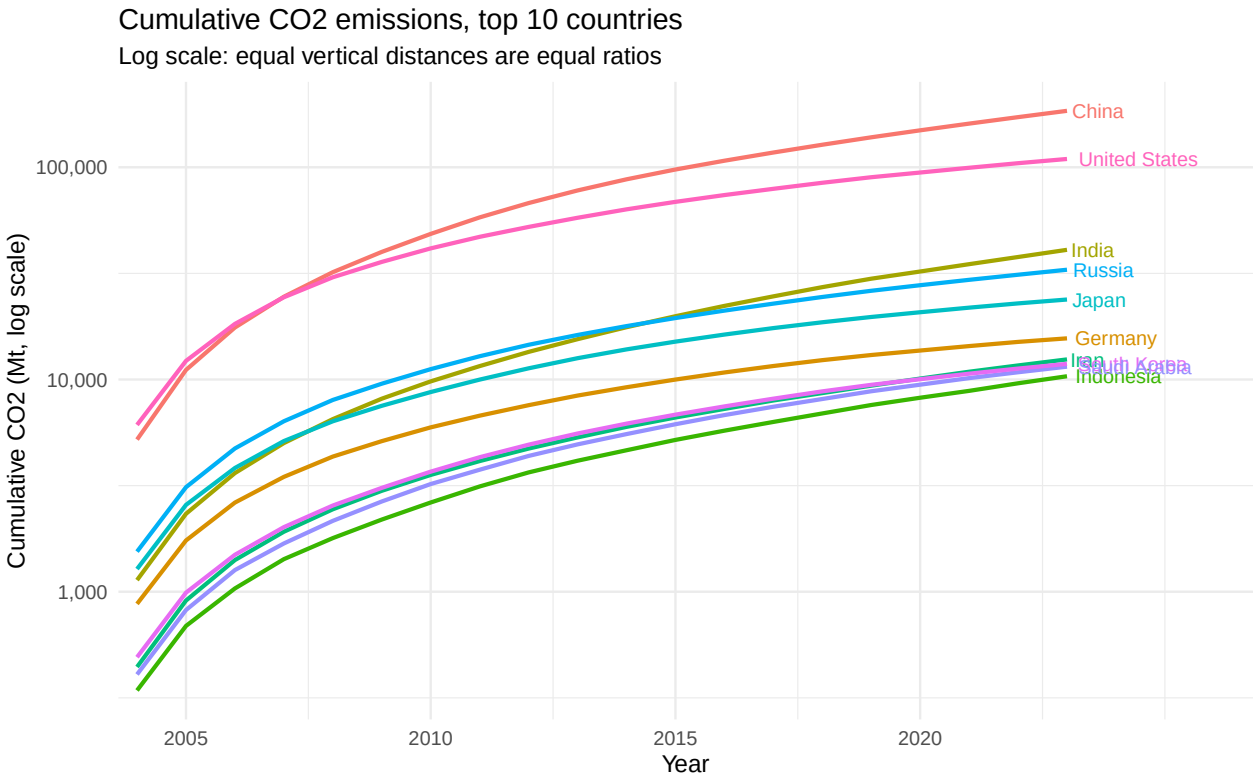
Region	Countries	CumCO2	CumOilCO2	CumRenew	CumConsCO2
Asia	39	345,927.5	81,256.4	40,331.9	323,929.9
North America	22	112,422.3	47,685.9	12,364.8	119,559.3
Europe	33	102,564.4	33,841.5	21,543.5	109,687.0
Africa	41	23,825.1	8,339.5	2,363.4	22,574.0
South America	10	20,729.6	12,344.6	13,963.1	20,920.3
Oceania	5	954.8	572.6	710.5	1,075.3

Observation: Asia accounts for 345,927.5 Mt, 57% of all cumulative emissions, and is the only region where consumption-based emissions (323,929.9 Mt) fall well below production, making it a net exporter of embodied carbon. Europe and North America are net importers: Europe consumed 109,687.0 Mt against 102,564.4 Mt produced. South America stands out for renewables, with 13,963.1 TWh against only 20,729.6 Mt of CO2.

2.6 Cumulative emissions over time, top 10 countries (log Y)

```
top10_series <- master %>% filter(ISO %in% top10total$ISO)
top10_labels <- top10_series %>% filter(Year == max(Year))

ggplot(top10_series, aes(Year, CumCO2, colour = Country)) +
  geom_line(linewidth = 0.9) +
  # Labels at the line ends replace a 10-colour legend lookup.
  geom_text(data = top10_labels, aes(label = Country), hjust = -0.1,
            size = 3, show.legend = FALSE) +
  scale_y_log10(labels = label_comma()) +
  scale_x_continuous(breaks = seq(2005, 2020, 5),
                     expand = expansion(mult = c(0.02, 0.2))) +
  labs(title = "Cumulative CO2 emissions, top 10 countries",
       subtitle = "Log scale: equal vertical distances are equal ratios",
       x = "Year", y = "Cumulative CO2 (Mt, log scale)") +
  theme(legend.position = "none")
```

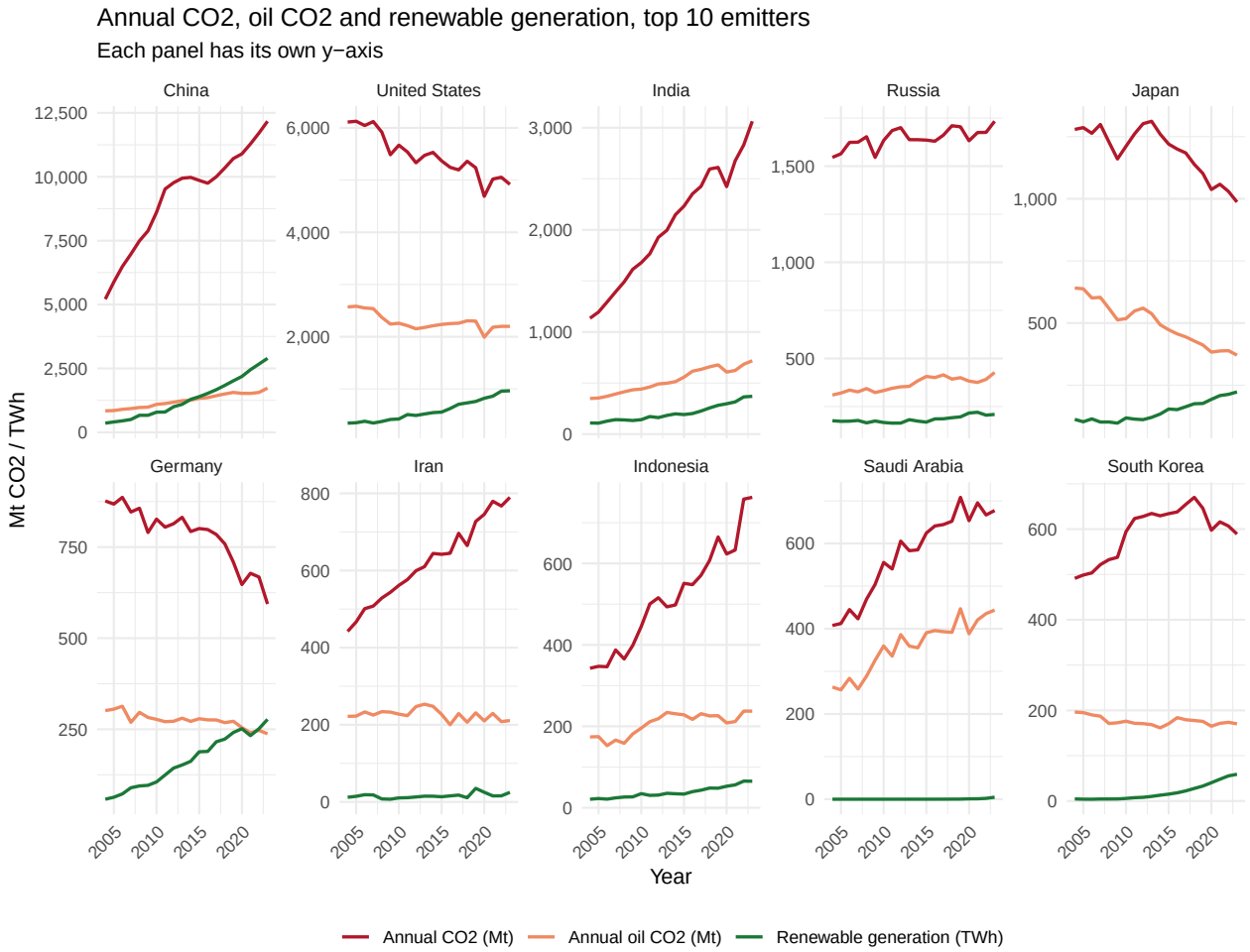


Observation: China (184,474.9 Mt) and the United States (109,445.5 Mt) are far ahead of the rest; China's line overtakes the United States' around 2007. India's line keeps a steeper slope than Japan's and Russia's, so it overtook both during the period. The steep rise in the first years is the running total starting in 2004, not a surge in emissions.

2.7 Faceted plot, one panel per country

```
top10_annual <- master %>%
  filter(ISO %in% top10total$ISO) %>%
  select(Country, Year, CO2, OilCO2, Renewables) %>%
  pivot_longer(c(CO2, OilCO2, Renewables), names_to = "Variable",
               values_to = "Value") %>%
  mutate(Variable = recode(Variable, CO2 = "Annual CO2 (Mt)",
                           OilCO2 = "Annual oil CO2 (Mt)",
                           Renewables = "Renewable generation (TWh)"),
         Country = fct_reorder(Country, Value, .fun = max, .desc = TRUE))

ggplot(top10_annual, aes(Year, Value, colour = Variable)) +
  geom_line(linewidth = 0.8) +
  facet_wrap(~ Country, ncol = 5, scales = "free_y") +
  scale_colour_manual(values = c("#B2182B", "#EF8A62", "#1B7837")) +
  scale_y_continuous(labels = label_comma()) +
  labs(title = "Annual CO2, oil CO2 and renewable generation, top 10 emitters",
       subtitle = "Each panel has its own y-axis",
       x = "Year", y = "Mt CO2 / TWh", colour = NULL) +
  theme(legend.position = "bottom",
        axis.text.x = element_text(angle = 45, hjust = 1))
```



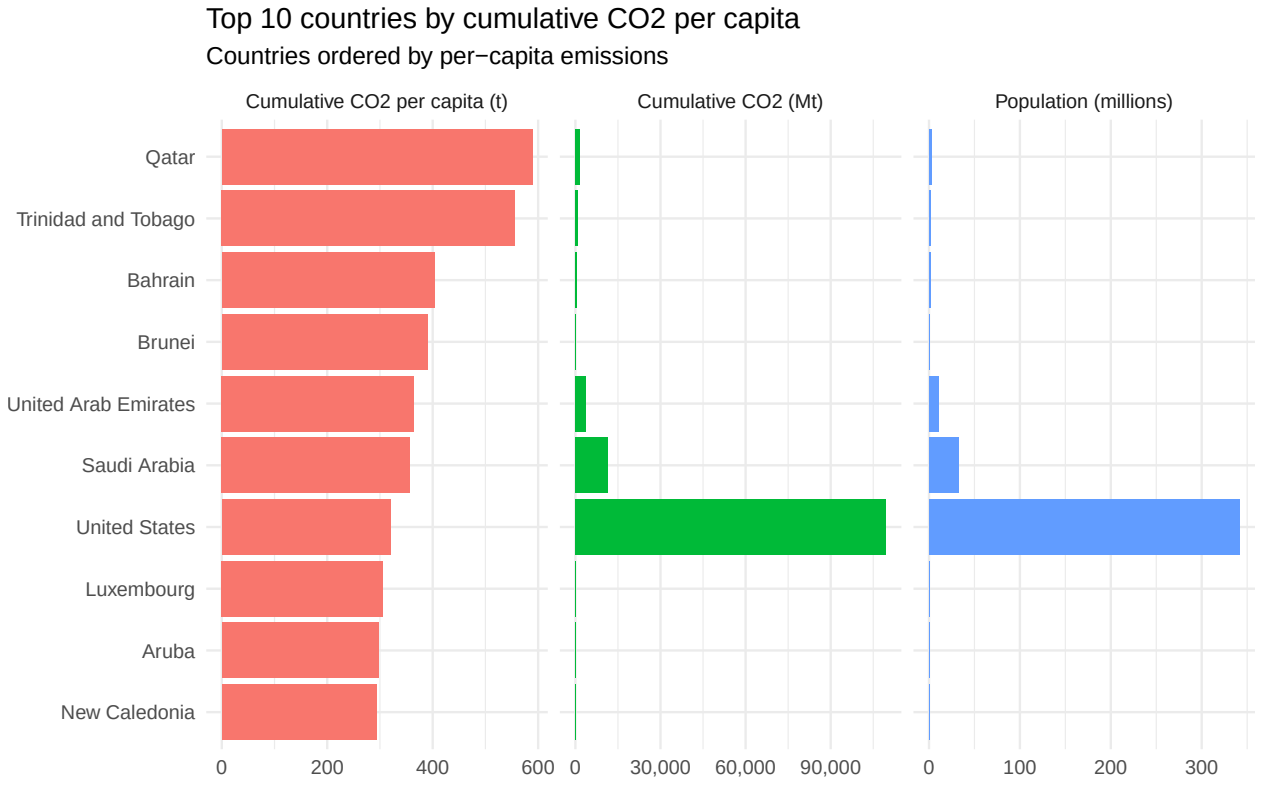
Observation: the panels split into two groups. In the United States, Japan and Germany annual CO2 declines while renewable generation rises (Germany's from 57.97 TWh in 2004 to 277.1 TWh in 2023). In China, India, Indonesia, Iran and Saudi Arabia emissions keep climbing; China's renewables grow fastest of all (357.43 to 2,894.22 TWh) but not fast enough to bend its emissions curve, and Saudi Arabia's renewable generation stays close to zero.

2.8 Top 10 by emissions per capita

```
percap_long <- top10percap %>%
  transmute(Country = fct_reorder(Country, CO2_PC),
            `Cumulative CO2 (Mt)` = CumCO2,
            `Population (millions)` = Population / 1e6,
            `Cumulative CO2 per capita (t)` = CO2_PC) %>%
  pivot_longer(-Country, names_to = "Measure", values_to = "Value") %>%
  mutate(Measure = factor(Measure, levels = c(
    "Cumulative CO2 per capita (t)", "Cumulative CO2 (Mt)",
    "Population (millions)")))

ggplot(percap_long, aes(Value, Country, fill = Measure)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ Measure, scales = "free_x") +
  scale_x_continuous(labels = label_comma()) +
```

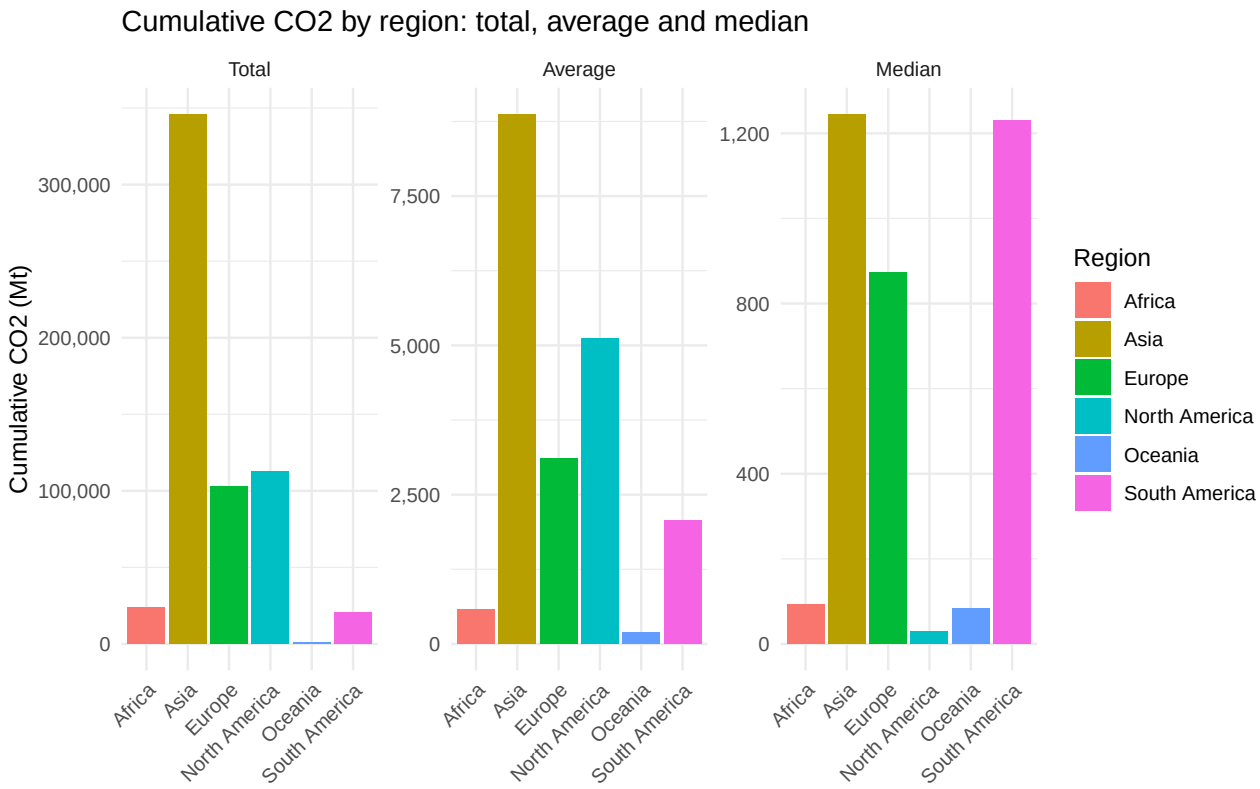
```
  labs(title = "Top 10 countries by cumulative CO2 per capita",
       subtitle = "Countries ordered by per-capita emissions",
       x = NULL, y = NULL)
```



2.9 Total, average and median cumulative emissions by region

```
region_stats <- lastYear_data %>%
  group_by(Region) %>%
  summarise(Total = sum(CumCO2), Average = mean(CumCO2),
            Median = median(CumCO2), .groups = "drop") %>%
  pivot_longer(-Region, names_to = "Statistic", values_to = "Value") %>%
  mutate(Statistic = factor(Statistic,
                            levels = c("Total", "Average", "Median")))

ggplot(region_stats, aes(Region, Value, fill = Region)) +
  geom_col() +
  facet_wrap(~ Statistic, scales = "free_y") +
  scale_y_continuous(labels = label_comma()) +
  labs(title = "Cumulative CO2 by region: total, average and median",
       x = NULL, y = "Cumulative CO2 (Mt)", fill = "Region") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Observation: comparing the three panels shows how skewed each region is. North America has the second-highest average (about 5,100 Mt) but one of the lowest medians, because the United States (109,445.5 Mt) dominates 22 mostly small countries. South America's median (about 1,230 Mt) almost matches Asia's despite a far smaller total, because its 10 countries are mostly mid-sized emitters with no single giant.

2.10 Top 2 countries per region by emissions per capita

```
top2_region <- lastYear_data %>%
  group_by(Region) %>%
  slice_max(CO2_PC, n = 2, with_ties = FALSE) %>%
  ungroup() %>%
  select(Region, Country, CO2_PC, CumCO2, Population) %>%
  arrange(Region, desc(CO2_PC))

kable(top2_region, digits = 1, format.args = list(big.mark = ","),
      caption = "Top 2 countries per region by cumulative CO2 per capita (t)")
```

Table 18: Top 2 countries per region by cumulative CO2 per capita (t)

Region	Country	CO2_PC	CumCO2	Population
Africa	Libya	158.0	1,141.4	7,223,804
Africa	South Africa	146.0	9,106.6	62,378,417
Asia	Qatar	588.8	1,703.1	2,892,466
Asia	Bahrain	403.5	618.8	1,533,460
Europe	Luxembourg	305.7	199.7	653,315

Region	Country	CO2_PC	CumCO2	Population
Europe	Estonia	236.5	319.3	1,350,093
North America	Trinidad and Tobago	556.3	832.2	1,495,923
North America	United States	320.5	109,445.5	341,534,041
Oceania	New Caledonia	293.4	84.2	287,130
Oceania	New Zealand	137.3	704.8	5,131,731
South America	Venezuela	102.7	2,896.4	28,213,015
South America	Argentina	79.9	3,627.6	45,407,904

2A. Counterintuitive finding

What did you expect, and why was that reasonable?

I expected the countries with the largest cumulative emissions to also lead the per-capita ranking, and expected any per-capita leaders outside that group to be oil-dominated economies. This was reasonable because a handful of large industrial economies dominate the totals (China alone emitted 184,474.9 Mt, 30% of the sample), and oil is the fuel most associated with high-consumption lifestyles and petro-states.

What does the data actually show? Quote a specific value, country or trend from your own plots.

Only two countries, the United States (320.5 t per person) and Saudi Arabia (357.3 t), appear in both `top10total` and `top10percap`. The per-capita leader is Qatar at 588.8 t per person, 4.5 times China's 129.4 t and 20 times India's 28.7 t, despite having just 2.9 million people and a cumulative total (1,703.1 Mt) under 1% of China's. Only 3 of the top 10 per-capita countries have more than 10 million people. The supporting chart also contradicts the second half of my expectation: Qatar, Bahrain, Brunei and the United Arab Emirates sit at the top-left in dark colours, meaning oil makes up under a quarter of their cumulative emissions, while the oil-dominated (yellow) points are mostly small island economies such as Aruba.

What real-world mechanism could explain the discrepancy?

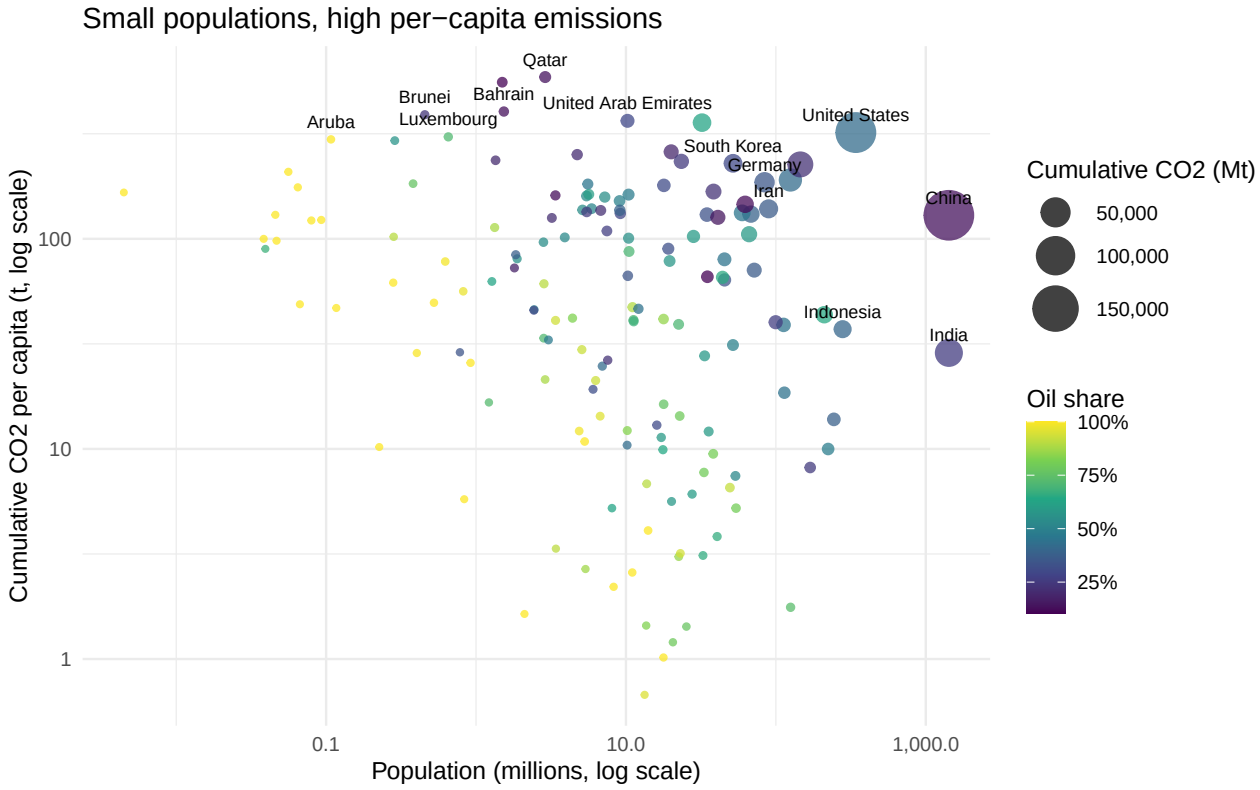
Two mechanisms combine. The first is population size: per-capita emissions divide a country's total by its population, so a country with fewer than 3 million people needs only a modest industrial base to post a very high figure. The second is energy resource endowment combined with production-based accounting. Qatar, Bahrain, Brunei and the UAE are natural-gas exporters: liquefying and processing gas, refining, petrochemicals and energy-intensive industries such as aluminium smelting, all powered by cheap domestic gas, emit CO2 inside their borders, even though much of the output is exported and consumed elsewhere. Production-based accounting assigns all of those emissions to a small resident population, which is why gas, not oil, drives the highest per-capita figures.

Supporting visualisation

```
# Population vs per-capita emissions, coloured by oil share: tests whether
# the per-capita leaders are small, oil- or gas-intensive economies rather
# than the large total emitters.
label_iso <- union(top10total$ISO, top10percap$ISO)

ggplot(lastYear_data, aes(Population / 1e6, CO2_PC)) +
  geom_point(aes(size = CumCO2, colour = OilShare), alpha = 0.75) +
  geom_text(data = filter(lastYear_data, ISO %in% label_iso),
            aes(label = Country), size = 2.8, vjust = -0.9,
            check_overlap = TRUE) +
```

```
  scale_x_log10(labels = label_comma()) +
  scale_y_log10(labels = label_comma()) +
  scale_colour_viridis_c(labels = label_percent()) +
  scale_size_continuous(labels = label_comma(), range = c(1, 10)) +
  labs(title = "Small populations, high per-capita emissions",
       x = "Population (millions, log scale)",
       y = "Cumulative CO2 per capita (t, log scale)",
       colour = "Oil share", size = "Cumulative CO2 (Mt)")
```



Caption (2–3 sentences): what does this chart show, and why did you choose this chart type?

Each point is one country in 2023, placed by population and cumulative CO2 per person, with bubble size showing its cumulative total and colour its oil share. The per-capita leaders cluster at the top-left (small populations), while the largest total emitters (the biggest bubbles, China and India) sit at the right with moderate per-capita values. I chose a scatter plot with log axes because population ranges from thousands to 1.4 billion and per-capita emissions from under 1 to 589 t, and size and colour let one chart test both parts of the explanation.

2B. Visualisation design critique

Visualisation chosen: Question 2.6, cumulative CO2 over time for the top 10 countries (log y-axis).

The first design decision was the log scale. Cumulative totals range from 10,360.2 Mt (Indonesia) to 184,474.9 Mt (China); on a log axis, equal vertical distances are equal ratios, so all ten growth paths are readable and comparable. The second was labelling each line at its end instead of using a legend, so a reader does not have to match ten colours. I rejected a linear y-axis: China and the United States would fill the plot while the

other eight countries collapsed into an indistinguishable band near zero, hiding the fact that India overtook Japan and Russia, which matters to a climate-policy audience tracking emerging emitters. The limitation is that the log scale shrinks gaps: China's lead over Indonesia is about 18 times, but the lines look close. A reader could also mistake the steep early rise for an emissions surge when it is the running total starting in 2004. I would mitigate both with an annotation stating the ratio and a subtitle explaining that the totals start in 2004.

Task 3: Data-driven modelling

3.1 Build `cor_data`

```
# Final-year record for each country (each country's own last year, in
# case a country's record ends early). Country names are kept as row
# names so the diagnostic plots in 3A label points by country.
cor_data <- master %>%
  group_by(ISO) %>%
  filter(Year == max(Year)) %>%
  ungroup() %>%
  select(Country, CumCO2, CumConsCO2, Population, GDP, GDPCapita) %>%
  column_to_rownames("Country") %>%
  as.data.frame()

dim(cor_data)
```

```
## [1] 150   5
```

```
head(cor_data)
```

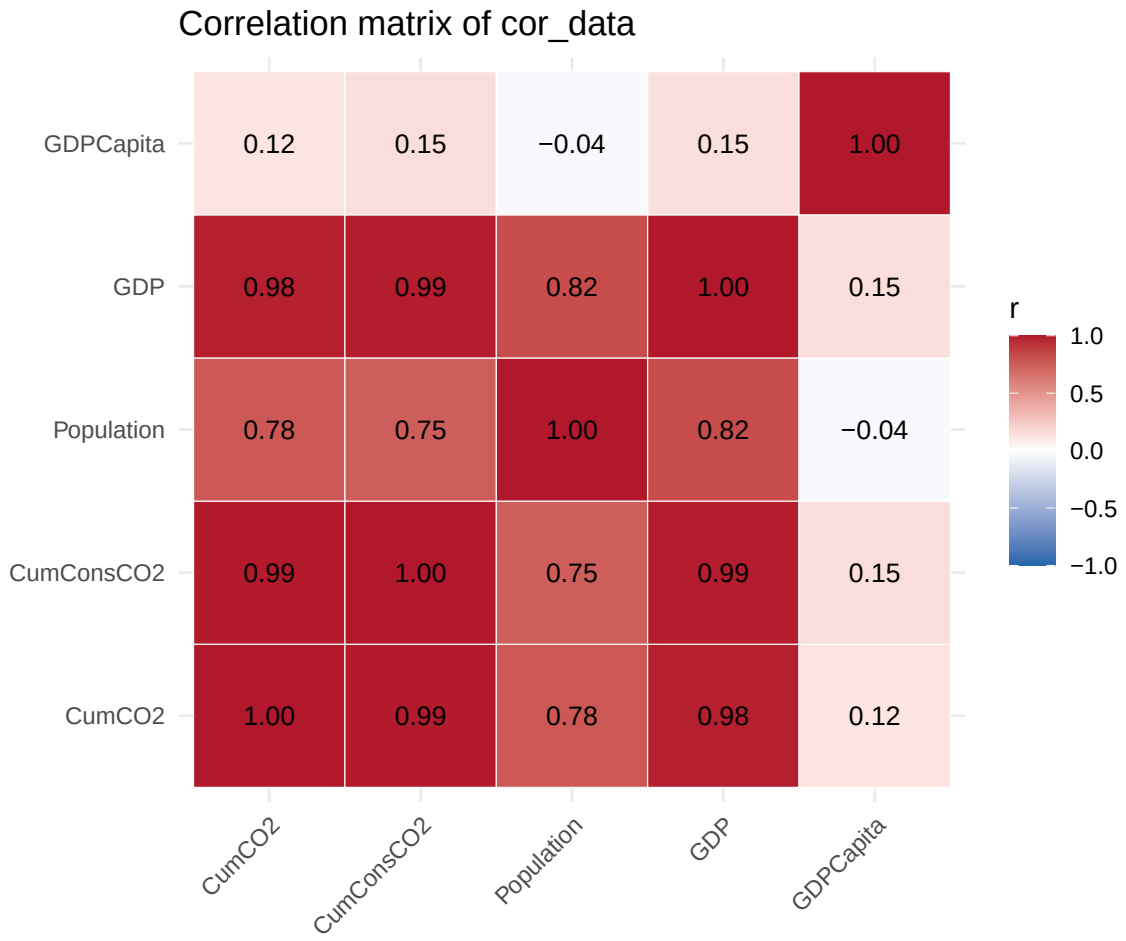
```
##                         CumCO2 CumConsCO2 Population        GDP GDPCapita
## Aruba                   32.053     32.053     107792   1194.874     11085
## Afghanistan            155.390    155.390   40578847  53303.000      1314
## Angola                 431.269    431.269   35635028 158346.000      4444
## Albania                 95.175    123.165    2827614  36171.000     12792
## Andorra                  9.749      9.749      79722   2263.068     28387
## United Arab Emirates  3732.660   4488.128   10242085 779702.000     76127
```

3.2 Correlation matrix

```
cor_matrix <- cor(cor_data)
round(cor_matrix, 2)
```

```
##            CumCO2 CumConsCO2 Population  GDP GDPCapita
## CumCO2       1.00       0.99       0.78 0.98      0.12
## CumConsCO2   0.99       1.00       0.75 0.99      0.15
## Population   0.78       0.75       1.00 0.82     -0.04
## GDP          0.98       0.99       0.82 1.00      0.15
## GDPCapita    0.12       0.15      -0.04 0.15      1.00
```

```
as.data.frame(as.table(cor_matrix)) %>%
  ggplot(aes(Var1, Var2, fill = Freq)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", Freq)), size = 3.5) +
  scale_fill_gradient2(low = "#2166AC", mid = "white", high = "#B2182B",
                       limits = c(-1, 1)) +
  labs(title = "Correlation matrix of cor_data", x = NULL, y = NULL,
       fill = "r") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
cor_gdp <- cor(cor_data$CumCO2, cor_data$GDP)
cor_gdp_pc <- cor(cor_data$CumCO2, cor_data$GDPCapita)
cat("cor(CumCO2, GDP)       =", round(cor_gdp, 2), "\n")
```

```
## cor(CumCO2, GDP)       = 0.98
```

```
cat("cor(CumCO2, GDPCapita) =", round(cor_gdp_pc, 2), "\n")
```

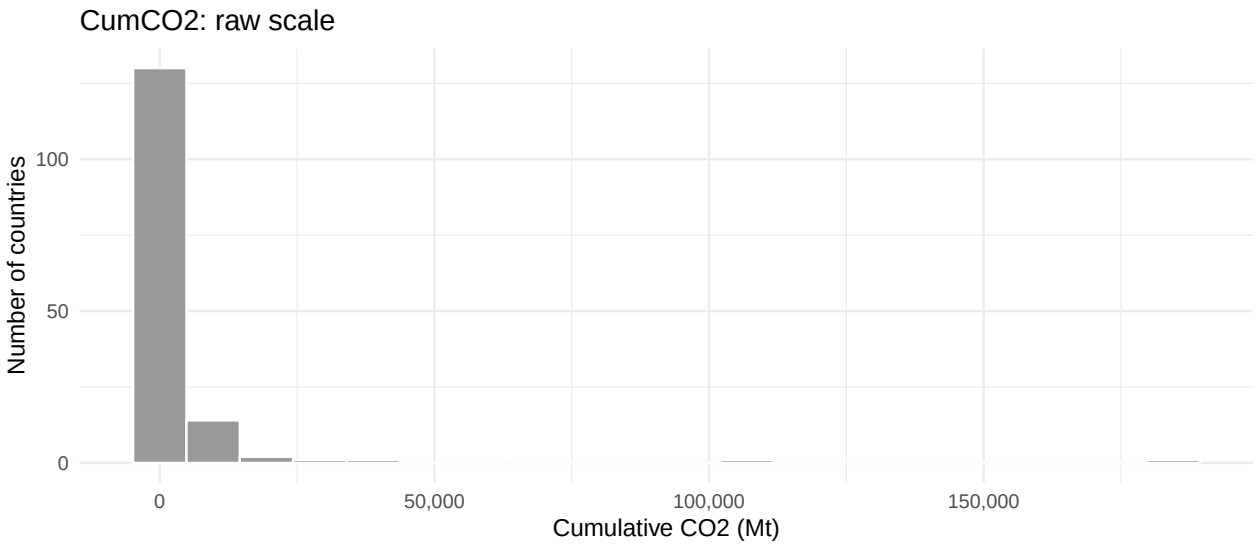
```
## cor(CumCO2, GDPCapita) = 0.12
```

Report `cor(CumCO2, GDP)` and `cor(CumCO2, GDPCapita)`. Why are those two numbers so different?

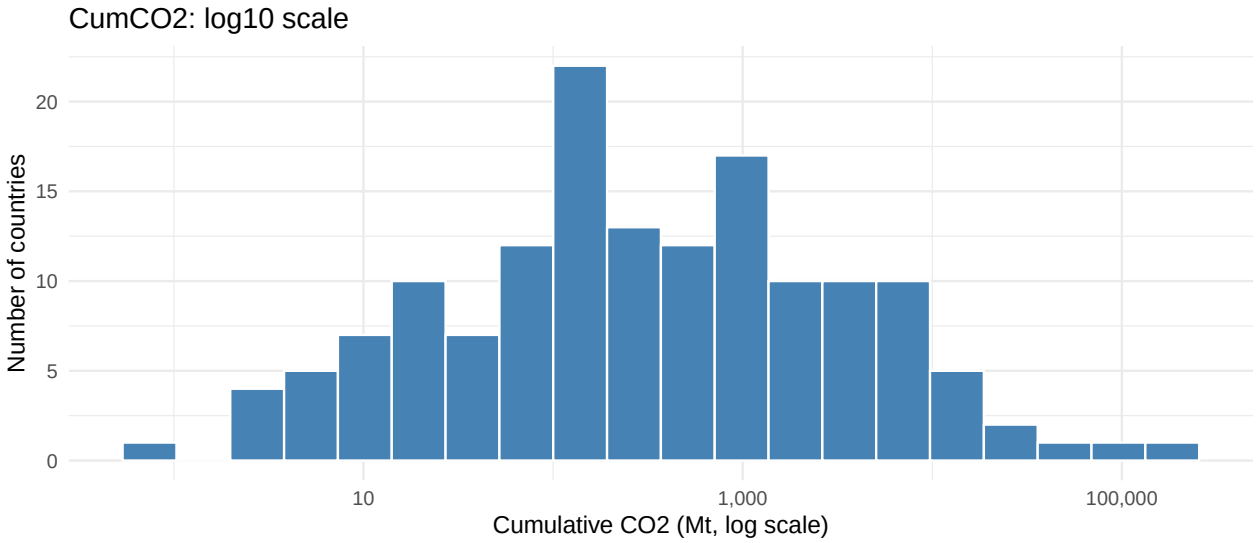
`cor(CumCO2, GDP)` is 0.98 and `cor(CumCO2, GDPCapita)` is 0.12. They differ because GDP and cumulative emissions are both measures of *size*: they grow with a country's population and total economic activity, so the same few giants (China, the United States, India) top both. GDP per capita removes size and measures wealth per person, which on its own says little about total emissions: rich but small countries such as Luxembourg and Singapore have small totals, while populous middle-income countries such as China (GDP per capita 18,921) and India (7,350) have the largest.

3.3 Distribution of `CumCO2`, with and without log

```r
ggplot(cor_data, aes(CumCO2)) +
  geom_histogram(bins = 20, fill = "grey60", colour = "white") +
  scale_x_continuous(labels = label_comma()) +
  labs(title = "CumCO2: raw scale", x = "Cumulative CO2 (Mt)",
       y = "Number of countries")
```




```r
ggplot(filter(cor_data, CumCO2 > 0), aes(CumCO2)) +
  geom_histogram(bins = 20, fill = "steelblue", colour = "white") +
  scale_x_log10(labels = label_comma()) +
  labs(title = "CumCO2: log10 scale", x = "Cumulative CO2 (Mt, log scale)",
       y = "Number of countries")
```



3.4 Train/test split

```r
## Required - do not remove or change this line:
set.seed(seed_from_id(params$studentID))

# Your 65/35 split goes below.
train_idx <- sample(seq_len(nrow(cor_data)),
                    size = round(0.65 * nrow(cor_data)))
train_data <- cor_data[train_idx, ]
test_data <- cor_data[-train_idx, ]

cat("Training rows:", nrow(train_data), "| Testing rows:", nrow(test_data),
    "\n")
```

```
## Training rows: 98 | Testing rows: 52
```

3.5 Linear model: CumCO2 ~ GDP

```r
model_gdp <- lm(CumCO2 ~ GDP, data = train_data)
summary(model_gdp)
```

```
##
## Call:
## lm(formula = CumCO2 ~ GDP, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22815.8   -190.2    740.0    884.3  19212.1
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.905e+02  4.274e+02  -2.083   0.0399 *
## GDP          6.162e-03  1.175e-04  52.450   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4066 on 96 degrees of freedom
## Multiple R-squared:  0.9663, Adjusted R-squared:  0.9659
## F-statistic:  2751 on 1 and 96 DF,  p-value: < 2.2e-16
```

```r
pred_gdp <- predict(model_gdp, newdata = test_data)
rmse_gdp <- Rmse(test_data$CumCO2, pred_gdp)
r2_gdp <- RSquared(test_data$CumCO2, pred_gdp)
cat("Test RMSE (GDP only):", round(rmse_gdp, 2), "Mt | Test R2:",
    round(r2_gdp, 3), "\n")
```

```
## Test RMSE (GDP only): 1719.84 Mt | Test R2: 0.812
```

3.6 Linear model: all predictors

```r
model_full <- lm(CumCO2 ~ CumConsCO2 + Population + GDP + GDPCapita,
                 data = train_data)
summary(model_full)
```

```
##
## Call:
## lm(formula = CumCO2 ~ CumConsCO2 + Population + GDP + GDPCapita,
##     data = train_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5759.4  -240.8    53.0   215.3  4223.3
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.542e+02  1.829e+02  -1.390    0.168
## CumConsCO2   1.786e+00  6.470e-02  27.605   <2e-16 ***
## Population   2.479e-05  1.772e-06  13.991   <2e-16 ***
## GDP         -5.204e-03  4.401e-04 -11.826   <2e-16 ***
## GDPCapita    5.963e-03  6.247e-03   0.955    0.342
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1270 on 93 degrees of freedom
## Multiple R-squared:  0.9968, Adjusted R-squared:  0.9967
## F-statistic:  7277 on 4 and 93 DF,  p-value: < 2.2e-16
```

```r
pred_full <- predict(model_full, newdata = test_data)
rmse_full <- Rmse(test_data$CumCO2, pred_full)
r2_full <- RSquared(test_data$CumCO2, pred_full)
cat("Test RMSE (all predictors):", round(rmse_full, 2), "Mt | Test R2:",
    round(r2_full, 3), "\n")
```

```
## Test RMSE (all predictors): 647.46 Mt | Test R2: 0.973
```

3.7 Interpret and compare the two models

When should the first be preferred, and when the second? If the full model has the higher test RMSE, explain why — that is not an error.

On the 52 test countries, the full model clearly outperforms the GDP-only model: its test RMSE is 647.46 Mt against 1,719.84 Mt, and its test R^2 is 0.973 against 0.812. In my split, the model with more predictors therefore does *not* have the higher test RMSE. Its advantage comes almost entirely from `CumConsCO2`, which correlates 0.99 with `CumCO2`. This is close to target leakage: consumption-based emissions are production-based emissions adjusted for trade, and for the 55 countries without consumption data `CumConsCO2` was set equal to `CumCO2` in Task 1.6, so the model partly learns an identity.

The GDP-only model should be preferred when the aim is explanation or when consumption data are unavailable. It is easy to interpret: each additional million dollars of GDP is associated with 0.00616 Mt (about 6,200 t) more cumulative CO2 since 2004. Its single coefficient is also stable, whereas the full model's coefficients are distorted by multicollinearity: GDP receives a *negative* coefficient (-0.0052 Mt per million dollars) and GDPCapita is not significant ($p = 0.342$), because GDP and `CumConsCO2` carry almost the same information ($r = 0.99$).

The full model is better for pure prediction when every predictor, including genuine consumption-based data, is available for the country being predicted.

Had the full model's test RMSE been higher, that would not have been an error: extra correlated predictors can overfit the 98 training countries and generalise worse, and with only 52 test countries a single large emitter can swing the RMSE.

3.8 Alternative models

```
predictors <- c("CumConsCO2", "Population", "GDP", "GDPCapita")
x_train <- as.matrix(train_data[, predictors])
x_test <- as.matrix(test_data[, predictors])
y_train <- train_data$CumCO2
y_test <- test_data$CumCO2

# Cross-validation folds are random; seeding again makes the lambda
# choice reproducible. The train/test split above is unaffected.
set.seed(seed_from_id(params$studentID))

# Ridge (alpha = 0) and lasso (alpha = 1); lambda chosen by 5-fold CV
# (5 rather than 10 folds because the training set is small).
ridge_cv <- cv.glmnet(x_train, y_train, alpha = 0, nfolds = 5)
lasso_cv <- cv.glmnet(x_train, y_train, alpha = 1, nfolds = 5)
pred_ridge <- as.numeric(predict(ridge_cv, newx = x_test,
                                 s = "lambda.min"))
pred_lasso <- as.numeric(predict(lasso_cv, newx = x_test,
                                 s = "lambda.min"))
coef(lasso_cv, s = "lambda.min")  # which predictors lasso keeps
```

```
## 5 x 1 sparse Matrix of class "dgCMatrix"
##                 lambda.min
## (Intercept) -1.926611e+02
## CumConsCO2   1.025043e+00
## Population   7.130379e-06
## GDP          .
## GDPCapita   -9.325435e-03
```

```
# Regression tree, pruned at the complexity with lowest CV error.
tree_fit <- rpart(CumCO2 ~ ., data = train_data, method = "anova",
                  control = rpart.control(minsplit = 6, cp = 0.001))
best_cp <- tree_fit$cptable[which.min(tree_fit$cptable[, "xerror"]), "CP"]
tree_pruned <- prune(tree_fit, cp = best_cp)
pred_tree <- predict(tree_pruned, newdata = test_data)

# Log-log linear model: addresses the skew seen in 3.3 and 3A.
# log1p() keeps zero values defined; predictions are back-transformed.
model_log <- lm(log1p(CumCO2) ~ log1p(CumConsCO2) + log1p(Population) +
                  log1p(GDP) + log1p(GDPCapita), data = train_data)
pred_log <- expm1(predict(model_log, newdata = test_data))

model_comparison <- tibble(
  Model = c("Linear (GDP only)", "Linear (all variables)",
```

```
            "Ridge (all variables)", "Lasso (all variables)",
            "Regression tree", "Log-log linear (all variables)"),
  RMSE = c(rmse_gdp, rmse_full, Rmse(y_test, pred_ridge),
           Rmse(y_test, pred_lasso), Rmse(y_test, pred_tree),
           Rmse(y_test, pred_log)),
  R_squared = c(r2_gdp, r2_full, RSquared(y_test, pred_ridge),
                RSquared(y_test, pred_lasso), RSquared(y_test, pred_tree),
                RSquared(y_test, pred_log))
) %>%
  arrange(RMSE)

kable(model_comparison, digits = 3, format.args = list(big.mark = ","),
      caption = "Test-set performance (RMSE in Mt CO2)")
```

Table 19: Test-set performance (RMSE in Mt CO2)

Model	RMSE	R_squared
Log-log linear (all variables)	577.657	0.979
Linear (all variables)	647.459	0.973
Lasso (all variables)	985.146	0.938
Ridge (all variables)	1,089.573	0.924
Linear (GDP only)	1,719.835	0.812
Regression tree	2,075.521	0.726

The log-log linear model performs best (test RMSE 577.66 Mt, R^2 0.979), because modelling every variable on a log scale removes the extreme right skew of `CumCO2` (3.3) so that a few giants no longer dominate the fit. The plain full linear model is second (647.46 Mt). Regularisation did not help: lasso (985.15 Mt) and ridge (1,089.57 Mt) shrink coefficients towards zero, which costs accuracy when there are only four predictors and a strong signal. Lasso's coefficients are informative, though: it dropped GDP entirely and kept `CumConsCO2` (coefficient 1.03), confirming that GDP adds nothing once consumption emissions are known. The regression tree is worst (2,075.52 Mt, R^2 0.726): its predictions are piecewise constant, and with only a handful of very large countries in 98 training rows it cannot represent the extreme values that dominate RMSE.

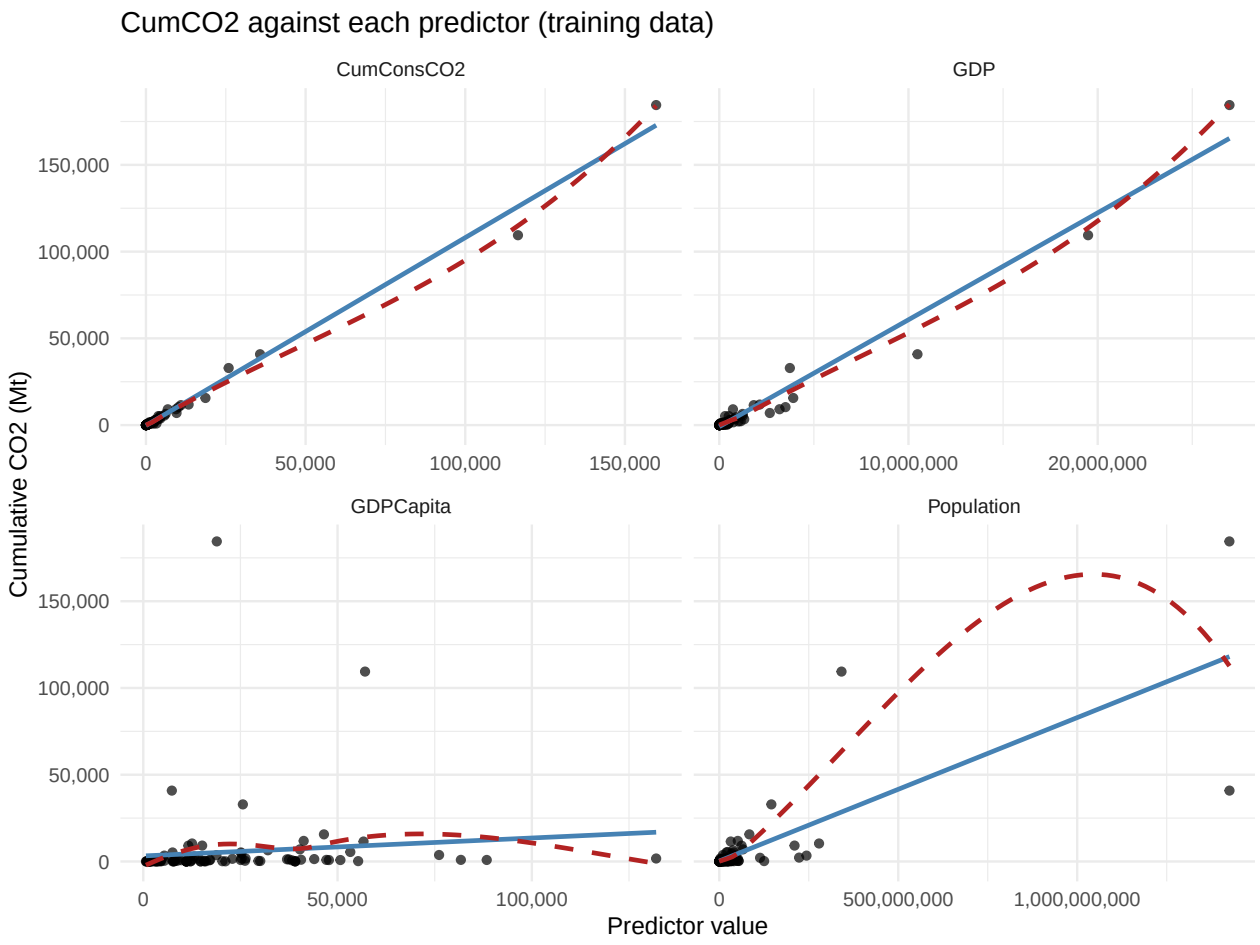
3A. Assumption checking

i. Linearity

```
train_long <- train_data %>%
  rownames_to_column("Country") %>%
  pivot_longer(all_of(predictors), names_to = "Predictor",
               values_to = "Value")

# Solid line: linear fit. Dashed line: flexible (loess) fit. Where the two
# diverge, a straight line is a poor description of the relationship.
ggplot(train_long, aes(Value, CumCO2)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, colour = "steelblue") +
```

```
  geom_smooth(method = "loess", se = FALSE, span = 0.9,
              colour = "firebrick", linetype = "dashed") +
  facet_wrap(~ Predictor, scales = "free_x") +
  scale_x_continuous(labels = label_comma()) +
  scale_y_continuous(labels = label_comma()) +
  labs(title = "CumCO2 against each predictor (training data)",
       x = "Predictor value", y = "Cumulative CO2 (Mt)")
```




```
# Pearson measures linear association, Spearman monotonic association; a
# large gap between them signals a non-linear relationship.
train_long %>%
  group_by(Predictor) %>%
  summarise(Pearson = cor(Value, CumCO2),
            Spearman = cor(Value, CumCO2, method = "spearman"),
            Gap = Spearman - Pearson, .groups = "drop") %>%
  kable(digits = 3)
```

Predictor	Pearson	Spearman	Gap
CumConsCO2	0.995	0.988	-0.007
GDP	0.983	0.947	-0.036
GDPCapita	0.105	0.533	0.429

Predictor	Pearson	Spearman	Gap
Population	0.776	0.687	-0.089

Does a linear relationship look reasonable? Which predictor departs from it most clearly?

A linear relationship looks reasonable for `CumConsCO2` and `GDP`: the linear and loess fits nearly coincide, and the Pearson correlations (0.995 and 0.983) match the Spearman rank correlations (0.988 and 0.947). This fit is largely driven by a few very large countries, though, with most countries bunched near the origin. `GDPCapita` departs from linearity most clearly: its linear correlation is only 0.105, but its rank correlation is 0.533, a gap of 0.429, so richer countries do tend to emit more but not in a straight line. `Population` also curves visibly, because China and India have similar populations but very different emissions.

ii. Distribution of the response

```
# Numerical support for the 3.3 histograms: a strongly right-skewed
# variable has a mean far above its median.
c(Mean = mean(train_data$CumCO2), Median = median(train_data$CumCO2))
```

```
##      Mean    Median
## 5312.5320  362.1947
```

```
shapiro.test(train_data$CumCO2)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  train_data$CumCO2
## W = 0.23617, p-value < 2.2e-16
```

```
shapiro.test(log1p(train_data$CumCO2))
```

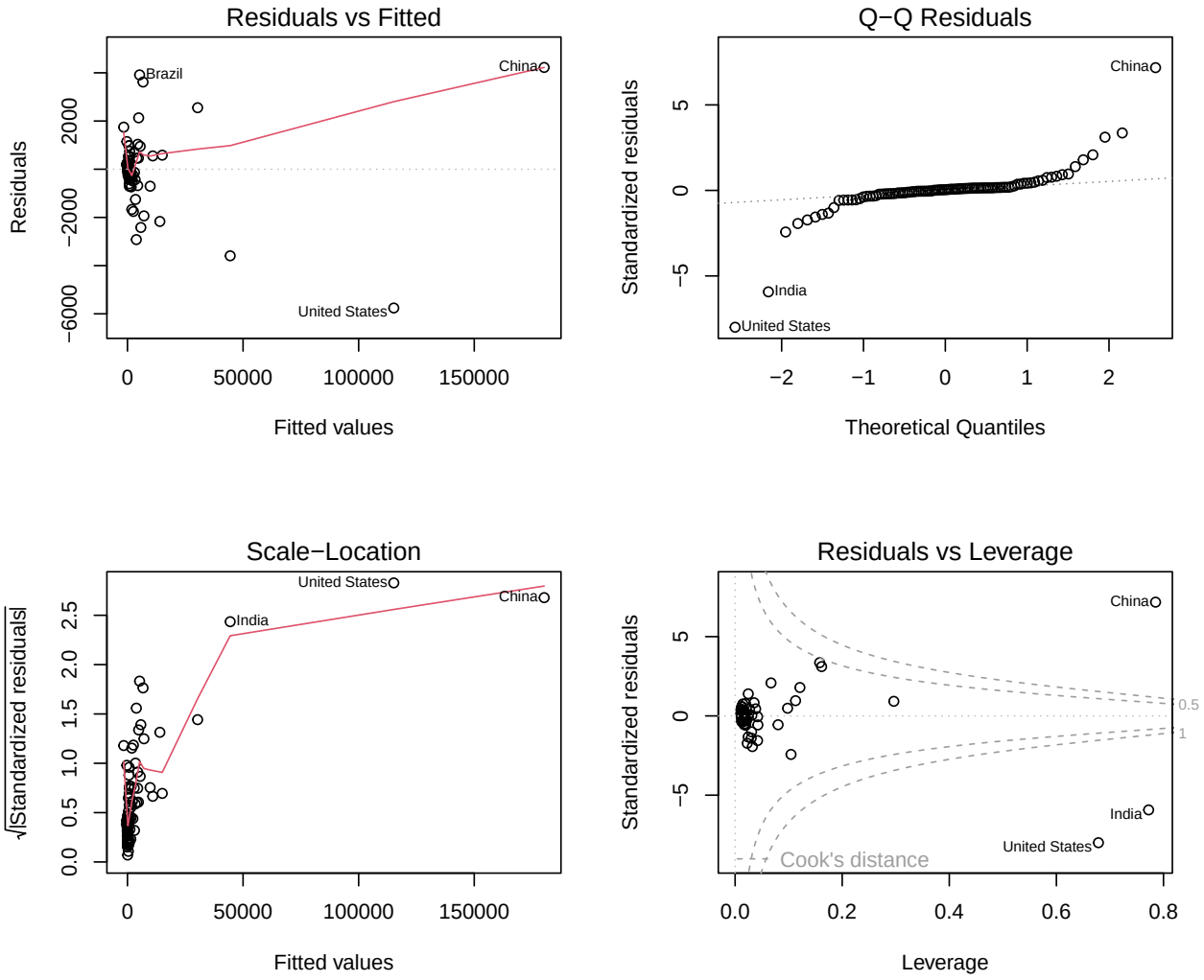
```
##
##  Shapiro-Wilk normality test
##
## data:  log1p(train_data$CumCO2)
## W = 0.99406, p-value = 0.9471
```

Using your plots from 3.3 — does raw `CumCO2` satisfy the normality assumption? Is a transformation warranted?

No. The raw histogram in 3.3 puts almost every country in the first bin, with a long right tail out to China (184,474.9 Mt). In the training data the mean (5,312.5 Mt) is almost 15 times the median (362.2 Mt), and the Shapiro–Wilk test rejects normality decisively ($W = 0.236$, $p < 2.2\text{e-}16$). On the log scale the histogram is roughly symmetric and the test no longer rejects normality ($W = 0.994$, $p = 0.947$), so a log transformation is warranted. Strictly, the assumption applies to the residuals rather than the response, but a response this skewed produces skewed residuals, as the Q–Q plot in iii confirms.

iii. Residual diagnostics

```
par(mfrow = c(2, 2))
plot(model_full, which = c(1, 2, 3, 5))
```




```
par(mfrow = c(1, 1))
```

Plot	What assumption it tests	Does your model satisfy it?
Residuals vs Fitted	Linearity: residuals should scatter randomly around 0 at every fitted value	No: the red line curves upward, with China's residual far above 0 and the United States' at about -5,800
Normal Q-Q	Normality of the residuals: points should follow the diagonal	No: both tails leave the line, with China above +5 and India and the United States below -5 standardised residuals
Scale-Location	Constant variance (homoscedasticity): the line should be flat	No: the line rises steeply from about 0.5 to 2.5, so the error spread grows with the size of the country

Plot	What assumption it tests	Does your model satisfy it?
Residuals vs Leverage	Influential observations: no point should lie beyond the Cook's distance contours	No: China (leverage 0.785), India (0.772) and the United States (0.678) lie beyond the Cook's distance 1 contour, against a threshold of 0.102

iv. Influential observations

```
# Leverage (hat value) and Cook's distance for each training country.
# Rule of thumb: leverage above 2p/n is high.
p <- length(coef(model_full))
n <- nrow(train_data)
influence_tbl <- tibble(Country = names(hatvalues(model_full)),
                        Leverage = hatvalues(model_full),
                        CooksD = cooks.distance(model_full)) %>%
  arrange(desc(Leverage))

cat("High-leverage threshold (2p/n):", round(2 * p / n, 3), "\n")
```

```
## High-leverage threshold (2p/n): 0.102
```

```
kable(head(influence_tbl, 5), digits = 3)
```

Country	Leverage	CooksD
China	0.785	37.646
India	0.772	23.846
United States	0.678	27.010
Qatar	0.296	0.071
Indonesia	0.162	0.373

```
high_lev_country <- influence_tbl$Country[1]
cat("High-leverage country:", high_lev_country, "\n")
```

```
## High-leverage country: China
```

```
model_full_drop <- lm(CumCO2 ~ CumConsCO2 + Population + GDP + GDPCapita,
                      data = train_data[rownames(train_data) !=
                                          high_lev_country, ])
rmse_full_drop <- Rmse(test_data$CumCO2,
                       predict(model_full_drop, newdata = test_data))
cat("Test RMSE with", high_lev_country, "removed:",
    round(rmse_full_drop, 2), "Mt (was", round(rmse_full, 2), "Mt)\n")
```

```
## Test RMSE with China removed: 546.71 Mt (was 647.46 Mt)
```

High-leverage country: China (leverage 0.785, Cook's distance 37.65)

New test RMSE without it: 546.71 Mt (down from 647.46 Mt)

Is removing it defensible? Two sentences.

Removing China is not defensible as data cleaning, because its values are genuine: it is the world's largest emitter (184,474.9 Mt), and a model that excludes it cannot describe the country that matters most for emissions policy. The lower test RMSE only shows that China was bending the fit for typical countries, a problem better addressed by the log transformation (577.66 Mt with China kept) than by deletion, unless China is explicitly modelled and reported separately.

v. Verdict

Is linear regression a valid choice here? If assumptions are violated, what does that mean in practice for your coefficients and predictions?

Linear regression on the raw scale is not a valid choice for inference here: all four diagnostics fail, and the fit is dominated by three countries. In practice, the standard errors and p-values in 3.6 cannot be trusted. The reported $p < 2e\text{-}16$ for GDP overstates certainty, and GDP's negative coefficient reflects multicollinearity with `CumConsCO2` ($r = 0.99$), not a real effect, so individual coefficients should not be interpreted. Predictions are also unreliable for typical countries: the test RMSE of 647.46 Mt is larger than the median training country's entire cumulative total (362.2 Mt), so relative errors for small countries are huge, and prediction intervals assuming constant variance would be far too narrow for large countries. The model is usable only as a rough predictor for large emitters. A log-log specification, which improved the test RMSE to 577.66 Mt and makes the response approximately normal ($W = 0.994$), is the better choice.

3B. Feature engineering critique

Approximately 300 words across the three parts.

i. Theoretical critique

State the physical pathway from GDP to emissions. Then explain why high correlation still does not license a causal claim: name at least two confounding or mediating variables GDP is proxying for. Reference your own correlation from 3.2, and contrast it with the `GDPCapita` correlation.

Producing goods and services requires energy; where that energy comes from coal, oil and gas, more output means more combustion and more CO2. Yet the 0.98 correlation does not make GDP a cause. Population is a confounder: bigger countries have bigger economies and bigger emissions (`Population` correlates 0.82 with GDP and 0.78 with `CumCO2`). Energy use and the carbon intensity of the energy mix are mediators: Norway's GDP per capita of 88,390 comes with 499.84 TWh of renewables per million people. The contrast with `GDPCapita` ($r = 0.12$) shows that once size is removed, wealth explains almost nothing, so most of GDP's 0.98 measures how big a country is.

ii. Proposed feature

Propose one engineered feature not in the dataset. Say exactly how you would construct it — what raw data, what calculation — and why it is more causally connected to emissions than GDP.

`CumFossilEnergy`: cumulative, carbon-weighted fossil-fuel consumption. From Our World in Data's energy dataset, take each country's annual coal, oil and gas primary-energy consumption (TWh), multiply each

fuel by its CO2 emission factor (coal emits roughly twice as much per unit of energy as gas), sum the three, and take a cumulative sum by country as in 1.8. It measures the combustion that physically releases CO2, capturing both mediators GDP only approximates, and separates a coal-powered economy from a hydro-powered one with the same GDP.

iii. Ethical risk

If this model allocated climate finance or set national reduction obligations, name one specific way it would be systematically unfair, and show that the unfairness comes from the structure of the model, not just "bias in the data".

Use your own data: name a country with large positive `NetImported` *and one with large negative* `NetImported`*, and build your argument around them.*

```
# Only countries with real consumption data: for the others NetImported is
# 0 by construction (Task 1.6), so they cannot illustrate the point.
covered_last <- lastYear_data %>% filter(!ConsImputed)

bind_rows(
  slice_max(covered_last, NetImported, n = 1) %>%
    mutate(Role = "Largest net importer"),
  slice_min(covered_last, NetImported, n = 1) %>%
    mutate(Role = "Largest net exporter")
) %>%
  select(Role, Country, NetImported, CumCO2, CumConsCO2, CO2_PC,
         ConsCO2_PC, GDP) %>%
  kable(digits = 1, format.args = list(big.mark = ","))
```

Role	Country	NetImported	CumCO2	CumConsCO2	CO2_PC	ConsCO2_PC	GDP
Largest net importer	United States	7,062.2	109,445.5	116,507.7	320.5	341.1	19,493,170
Largest net exporter	China	-24,691.0	184,474.9	159,783.8	129.4	112.1	26,966,017

```
# What the GDP-only model (3.5) predicts for the same two countries, and
# how far each country's actual production sits from that prediction.
covered_last %>%
  filter(Country %in% c("United States", "China")) %>%
  mutate(PredictedCumCO2 = predict(model_gdp, newdata = .),
         ProductionGap = CumCO2 - PredictedCumCO2,
         ConsumptionGap = CumConsCO2 - PredictedCumCO2) %>%
  select(Country, GDP, CumCO2, CumConsCO2, PredictedCumCO2,
         ProductionGap, ConsumptionGap) %>%
  kable(digits = 1, format.args = list(big.mark = ","))
```

Country	GDP	CumCO2	CumConsCO2	PredictedCumCO2	ProductionGap	ConsumptionGap
China	26,966,017	184,474.9	159,783.8	165,262.8	19,212.1	-5,479.0
United States	19,493,170	109,445.5	116,507.7	119,218.3	-9,772.8	-2,710.6

The unfairness is built into the model: its target is production-based emissions and its single GDP slope treats every dollar as equally carbon-intensive, so emissions embodied in trade are charged to the producer. The

United States, the largest net importer (`NetImported` = +7,062.2 Mt), consumed 116,507.7 Mt but produced 109,445.5 Mt; China, the largest net exporter (-24,691.0 Mt), produced 184,474.9 Mt but consumed 159,783.8 Mt. Against the GDP-only predictions (about 119,200 Mt and 165,300 Mt), China's production sits about 19,200 Mt above its prediction and the United States' about 9,800 Mt below. Obligations set from this gap would brand China an over-emitter and the United States an under-emitter, yet on a consumption basis both sit below their predictions. Any model of this structure rewards importing manufactured goods and penalises making them, however clean the data.

Task 4: Insights

Narrative thesis

One sentence: the single story a reader should take away from your dashboard.

YOUR ANSWER

Dashboard sketch

Embed your photographed hand-drawn sketch below. This is mandatory — without it you lose all design marks.

4.1 Data augmentation

How would you augment this dataset for a comprehensive view of the transition? What sources, and what would each add?

YOUR ANSWER

4.2 Key insights

Which insights would you highlight for a public audience? What figures would you use, and why is each important?

YOUR ANSWER

4.3 Design elements

Layout, colour, typography, placement.

YOUR ANSWER

4.4 Audience consideration

How does the design serve a public audience — accessible, understandable, actionable?

YOUR ANSWER

4.5 Additional questions

One relevant metric or visualisation you deliberately omitted, and why omitting it serves clarity.

YOUR ANSWER

One way your dashboard could be misread, and how design mitigates it.

YOUR ANSWER

Does your headline figure use total or per-capita emissions? Defend that choice against the obvious objection.

YOUR ANSWER

Conclusions

What are the main things your analysis shows? Keep it grounded in your own numbers.

YOUR ANSWER

Section 5: Verification

5.1 Results Summary Table

```
# Prints every value needed for the table below, straight from the
# objects computed above, so the table can be filled in without retyping
# numbers by hand from different sections.
neg_label <- if (nrow(negative_rows) == 0) "none" else
  paste(paste(negative_rows$Country, negative_rows$Year), collapse = "; ")

cat(" 1. Fingerprint:", my$fingerprint, "\n",
    "2. First and last year:", min(master$Year), "-", max(master$Year),
    "\n",
    "3. Countries in Emissions.csv:", n_countries, "\n",
    "4. dim(master):", paste(dim(master), collapse = " x "), "\n",
    "5. Duplicate rows removed:", n_dup_removed, "\n",
    "6. Negative-emissions rows:", neg_label, "\n",
    "7. Countries with no consumption data:", n_no_cons, "\n",
    "8. Highest-CO2 year and total:", peak_year$Year, "/",
    round(peak_year$TotalCO2, 1), "Mt\n",
    "9. Top cumulative / per capita:", top10total$Country[1], "/",
    top10percap$Country[1], "\n",
    "10. cor(CumCO2, GDP) / cor(CumCO2, GDPCapita):", round(cor_gdp, 2),
    "/", round(cor_gdp_pc, 2), "\n",
    "11. Test RMSE, GDP-only model:", round(rmse_gdp, 2), "\n",
    "12. Test RMSE, full model:", round(rmse_full, 2), "\n",
    "13. High-leverage country:", high_lev_country, "\n")
```

```
##  1. Fingerprint: EM-606669-0798-6584
##  2. First and last year: 2004 - 2023
##  3. Countries in Emissions.csv: 150
##  4. dim(master): 3000 x 26
##  5. Duplicate rows removed: 5
##  6. Negative-emissions rows: Equatorial Guinea 2010; Sri Lanka 2012; Sudan 2018; Sweden 2022
##  7. Countries with no consumption data: 55
##  8. Highest-CO2 year and total: 2023 / 34067.6 Mt
##  9. Top cumulative / per capita: China / Qatar
##  10. cor(CumCO2, GDP) / cor(CumCO2, GDPCapita): 0.98 / 0.12
##  11. Test RMSE, GDP-only model: 1719.84
##  12. Test RMSE, full model: 647.46
##  13. High-leverage country: China
```

Fill every cell by hand from your own output. If you could not compute one, write `NA` *and say why in a footnote below the table.*

#	Quantity	Your value
1	Dataset fingerprint (printed by the setup chunk)	EM-606669-0798-6584
2	First and last year in your dataset	2004 and 2023
3	Number of countries in your `Emissions.csv`	150

#	Quantity	Your value
4	`dim()` of your final merged master data frame	3000 x 26
5	Number of duplicate rows you removed in Task 1	5
6	Countries and years of the negative-emissions rows (1A ii)	Equatorial Guinea 2010; Sri Lanka 2012; Sudan 2018; Sweden 2022
7	Number of countries with no consumption-based data (1A iii)	55
8	Year with the highest total CO2, and that total (Task 2.1)	2023, 34,067.6 Mt
9	Top country by cumulative emissions, and by emissions per capita	China (cumulative); Qatar (per capita)
10	`cor(CumCO2, GDP)` and `cor(CumCO2, GDPCapita)` (2 dp)	0.98 and 0.12
11	Test RMSE of the GDP-only model (Task 3.5)	1,719.84 Mt
12	Test RMSE of the full model (Task 3.6)	647.46 Mt
13	The high-leverage country you identified (3A iv)	China

5.2 Reproducibility appendix

Run this chunk unchanged. Do not edit it.

```
cat("Student ID  :", params$studentID, "\n")
```

```
## Student ID  : u3284342
```

```
cat("Fingerprint :", my$fingerprint, "\n")
```

```
## Fingerprint : EM-606669-0798-6584
```

```
cat("Window      :", paste(my$window, collapse = " - "), "\n")
```

```
## Window      : 2004 - 2023
```

```
cat("Knitted at  :", format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"), "\n")
```

```
## Knitted at  : 2026-11-10 13:31:24 AEDT
```

```
cat("Chunks run  :", length(knitr::all_labels()), "\n\n")
```

```
## Chunks run  : 44
```

```r
print(sessionInfo()$R.version$version.string)
```

```
## [1] "R version 4.6.1 (2026-06-24)"
```

Appendix: Decision Log

*Minimum five entries. Write each one **at the time you make the decision**, not at the end. Every entry must reference something specific from your own analysis — a variable name, a country, a line of code, an output value. Generic entries receive zero marks. Add rows freely; more than five is viewed favourably.*

#	Task & step	Decision made	Alternative considered	Why I chose this	Impact on results
1					
2					
3					
4					
5					

Declaration

I confirm that the dataset analysed in this submission was generated from my own student ID, that the values reported in Section 5.1 are the values produced by the code in this document, and that all explanations and interpretations are my own work.

Student ID: _____ **Date:** _____